


# End-of-Degree Project (Business)

## In-work poverty: the case of Spain

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## Title in Spanish

Pobreza laboral: el caso de España

## Abstract

Using the 2024 wave of the EU-SILC for Spain (income year 2023) and Eurostat's harmonised in-work-poverty definition, this dissertation finds that 11.3 % of Spanish workers live in households below the 60 %-median-income line. Weighted descriptives and stepwise logistic regressions reveal that earning a low hourly wage raises the odds of poverty almost five-fold; temporary contracts, involuntary part-time work and vulnerable self-employment further intensify the risk, while single-parent families and very-low-work-intensity households face odds six and twenty times higher, respectively. Young adults, migrants and workers with low educational attainment are disproportionately affected, whereas tertiary education and home-ownership offer protection. The evidence indicates that a comprehensive policy mix, such as higher minimum wages, stronger collective bargaining, incentives for full-time stable jobs, affordable childcare and better-targeted family transfers, is required if employment is to regain its role as a reliable route out of poverty in Spain.

## Keywords

in-work poverty, household structure, logistic regression, Spain, 2024

## Resumen

Analizamos la pobreza laboral en España utilizando la Encuesta de Condiciones de Vida 2024 (ingresos 2023) y la definición armonizada de Eurostat para "trabajador pobre". La incidencia alcanza el 11,3 % de las personas ocupadas. Mediante modelos logísticos escalonados se identifican los determinantes clave: percibir un salario bajo multiplica por cinco la probabilidad de pobreza; la temporalidad, el trabajo a tiempo parcial y el autoempleo vulnerable agravan el riesgo, mientras que los hogares monoparentales y con muy baja intensidad laboral registran probabilidades seis y veinte veces mayores, respectivamente. La vulnerabilidad se concentra en jóvenes, migrantes y personas con baja educación, en tanto que la educación terciaria y la propiedad de la vivienda protegen frente a la pobreza. Se concluye que un paquete de políticas que combine un salario mínimo robusto, fomento del empleo estable y a tiempo completo, cuidados asequibles y transferencias mejor orientadas a familias con hijos resulta imprescindible para que el empleo vuelva a ser una vía efectiva de inclusión social en España.

## Palabras clave

pobreza laboral, estructura del hogar, regresión logística, España, 2024

# 1. Introduction

In-work poverty refers to the condition of being employed yet living in a household with income below the poverty threshold. In official European statistics, it is usually captured by the in-work at-risk-of-poverty rate, which measures the share of working individuals whose equivalised disposable household income falls below a given poverty line (typically 60% of the national median) (Latta & Peña-Casas, 2004).

The scope of in-work poverty analysis therefore extends beyond individual earnings, encompassing household demographics and the role of taxes and transfers. It is important to distinguish this from low pay or earning poverty at the individual level: not all low-wage workers are in poor households, and conversely, not all workers in poor households have low personal wages (their poverty may stem from insufficient working hours or many dependents). In other words, in-work poverty is a multidimensional phenomenon influenced by both labour market outcomes (wages, hours, job stability) and household circumstances (number of earners, family size, etc.).

In-work poverty undermines the idea that employment guarantees prosperity and creates a series of mutually reinforcing problems: it depresses productivity and human-capital investment, locks workers into low-value-added, high-turnover jobs, strains public finances through greater reliance on tax credits and social transfers, and can trap people in poverty over time. Socially, it widens income gaps, concentrates disadvantage among young people, migrants, women and the low-skilled, harms physical and mental well-being (including that of children), and corrodes social cohesion by eroding trust in the "work-pays" social contract. These intertwined economic and social costs explain why reducing in-work poverty requires coordinated labour-market, fiscal and social-policy action.

Given the trends and concerns outlined above, this dissertation sets out to explore in-work poverty in Spain through a structured inquiry. The general objective of the study is to provide a comprehensive analysis of the incidence and drivers of in-work poverty in Spain. This broad goal can be broken down into several specific objectives and research questions, which guide the investigation and structure of the dissertation.

The objectives of the study can be outlined as follows:

- **Objective 1**: To quantify and describe the incidence of in-work poverty in Spain, using the latest available data. This includes examining the current state of the in-work poverty rate in Spain and broadly comparing its situation with European averages or specific countries.

- **Objective 2**: To identify and analyse the determinants of in-work poverty at both the individual level (e.g., low wages, unstable jobs, part-time work) and the household level (e.g., single-earner households, large families) in Spain. This objective will likely involve examining empirical evidence or studies (such as regression analyses or decomposition studies by authors like Lanau & Lozano, 2024, or Tejero et al., 2024) that shed light on why certain workers become or remain poor.

By pursuing these objectives, the dissertation aims to answer the research questions in a systematic way. The overall ambition is to contribute to an informed understanding of in-work poverty in Spain – what are the socioeconomic factors associated with it and who is affected. Each chapter of the dissertation will tackle different aspects of these objectives, as outlined below.

This dissertation is structured into five chapters, each of which addresses a specific dimension of the research questions posed.

- The first chapter, Introduction, has provided the general background of the study, defining the concept of in-work poverty and its scope, highlighting the relevance of analysing this issue both in Spain and in the broader European context, and discussing the economic and social implications of the persistence of poverty among the employed population.
- The second chapter, entitled Literature review and theoretical framework, is devoted to the systematic examination of the existing academic literature on in-work poverty, with a particular focus on its main determinants and underlying mechanisms.
- The third chapter, Data and methodology, presents the empirical strategy employed in the dissertation, justifying the focus on Spain and the use of recent data from the Encuesta de Condiciones de Vida (ECV 2024), the Spanish implementation of the European Union Statistics on Income and Living Conditions (EU-SILC).
- The fourth chapter, Empirical results, constitutes the core of the dissertation's analytical contribution.
- The fifth and final chapter, Conclusions, summarises the main findings of the dissertation and provides answers to the research questions formulated in the introductory chapter.

# 2. Literature review and theoretical framework

## 2.1. The role of the labour market in the emergence of in-work poverty

A core element in understanding the persistence of in-work poverty across Europe, and particularly in Spain, lies in the structural features of the labour market. Empirical research consistently shows that not all forms of employment provide sufficient income to avoid poverty, especially when job quality is poor, work intensity is low, or employment arrangements are unstable (Nolan & Marx, 2014; Peña-Casas & Latta, 2004; Eurofound, 2017). In this regard, the segmentation of the labour market into a protected core and a precarious periphery plays a critical role in shaping both the likelihood of being working poor and the depth of income deprivation experienced by this group. The Spanish case has been extensively analysed as an example of this dualism, where a large share of new employment opportunities, particularly after the financial crisis, have been concentrated in low-wage and insecure forms of work (Lanau & Lozano, 2024; Ibáñez, Tejero & López-Rodríguez, 2024). This section explores how specific labour market characteristics contribute to in-work poverty, with particular attention to temporary contracts, part-time work, low wages, vulnerable self-employment, and emerging forms of structural precariousness.

### *2.1.1. Temporary contacts, part-time work and low wages*

The first set of labour market factors that explain the emergence and persistence of in-work poverty relates to employment conditions that directly constrain earnings capacity. Temporary contracts, involuntary part-time work, and low hourly wages have been consistently identified as key risk factors for poverty among workers (Marx & Nolan, 2014; Eurofound, 2017). These features are especially relevant in Spain, where the incidence of precarious work arrangements has historically exceeded the European average.

Temporary employment represents one of the most significant drivers of in-work poverty in the Spanish labour market. Spain has long maintained one of the highest rates of temporary contracts in the European Union, with peaks above 25% of salaried employment during several years of the last two decades (Eurostat, 2021). Temporary workers tend to have lower hourly wages, fewer hours worked per year, and limited access to social security benefits compared to permanent employees (Eurofound, 2017). Moreover, the instability associated with contract discontinuity and the prevalence of short-term contracts often lead to periods of unemployment between contracts, which reduces annual household income even when the individual is formally employed during part of the year (Lanau & Lozano, 2024). The recent labour market reform introduced in Spain in 2021-2022, aimed at reducing the excessive use of temporary hiring, may partially alleviate this structural vulnerability, but its full impact on in-work poverty remains to be evaluated.

Part-time work constitutes another important element in the relationship between employment and poverty risk. Although part-time employment is not inherently associated with poverty—particularly when chosen voluntarily or when combined with other sources of household income—it becomes a significant risk factor when it is involuntary or marginal (Horemans & Marx, 2013). In Spain, the rise of involuntary part-time work, especially among women, young people, and migrant workers, has been notable in the aftermath of the Great Recession. According to INE data (2021), a considerable share of part-time workers in Spain declare that they would prefer full-time employment but are unable to find it. This underemployment leads to reduced earnings and contributes to low work intensity at the household level, which is strongly associated with poverty status (Eurofound, 2017). Nevertheless, the relationship between part-time work and in-work poverty is mediated by household composition: secondary earners in dual-income households may hold part-time jobs without risk of poverty, whereas primary earners working part-time are much more exposed to income insufficiency (Marx & Nolan, 2014).

Low wages, in turn, remain one of the most direct mechanisms through which employment fails to ensure adequate living standards. Spain exhibits a wide dispersion in wage levels, with a significant segment of workers concentrated around the minimum wage, particularly in sectors such as agriculture, hospitality, domestic work, and retail (Gómez-Álvarez & Gómez, 2017). The link between low hourly pay and in-work poverty has been widely documented: although not all low-paid workers are poor—since many live in multi-earner households—the probability of being working poor is substantially higher among low-wage earners than among higher-paid workers (Peña-Casas & Latta, 2004; Lohmann, 2018). In Spain, sluggish wage growth during the recovery period following the 2008–2014 crisis, combined with the expansion of low-productivity service jobs, reinforced this risk.

The role of minimum wage policy is particularly relevant in this context. For much of the last decade, Spain's statutory minimum wage was relatively low, standing at approximately €750 per month in 2016 (paid in 14 instalments). A significant policy shift occurred in 2019, with a 22% increase in the minimum wage, followed by additional rises that brought the SMI to €1,080 by 2023. These increases were explicitly justified on grounds of reducing wage inequality and tackling in-work poverty (OECD, 2020). Early empirical analyses suggest that these policy interventions have contributed to raising earnings at the bottom of the wage distribution without significant negative effects on employment levels (ILO, 2020). However, as noted by several authors (Hijzen, Montenegro & Pessoa, 2023; Ibáñez et al., 2024), raising the minimum wage alone cannot fully eliminate in-work poverty,

particularly in cases where poverty is driven not only by hourly pay but also by low working hours, temporary employment, or high household dependency ratios.

In sum, temporary contracts, involuntary part-time work, and low wages represent interrelated mechanisms through which labour market dynamics generate and sustain in-work poverty. Their combined effect is particularly detrimental for single earners and for workers in households with dependents, highlighting the need to understand these employment conditions within the broader context of household composition and welfare state interventions.

### *2.1.2. Vulnerable self-employment and structural precariousness*

Beyond standard employment relations, a growing body of literature has highlighted the role of vulnerable self-employment and structural precariousness as significant contributors to in-work poverty (Eurofound, 2017; Mussida & Sciulli, 2025). Self-employment, particularly in the form of own-account work without employees, is frequently associated with unstable income flows, limited access to social protection, and exposure to market fluctuations that undermine economic security.

In the Spanish labour market, self-employment (autoempleo) remains prevalent in sectors characterised by low productivity and low earnings, such as agriculture, construction, retail trade, and personal services. According to Eurostat data (2021), self-employed individuals in Spain exhibit at-risk-of-poverty rates that are consistently higher than those of salaried employees, often by a factor of two. This pattern is not unique to Spain but is particularly pronounced in contexts where self-employment is less a voluntary entrepreneurial choice and more a necessity-driven alternative to unemployment (Peña-Casas & Latta, 2004; Ibáñez et al., 2024).

The concept of vulnerable self-employment encompasses those own-account workers who lack economic autonomy, such as those engaged in economically dependent self-employment or "false" self-employment arrangements. These workers often depend on a single client or operate within subcontracting chains where bargaining power is minimal. The expansion of the platform economy and the gig sector in recent years has further amplified these dynamics. Platform-based work, such as food delivery or ride-hailing services, typically classifies workers as independent contractors while offering few guarantees in terms of minimum earnings, working time regularity, or social insurance coverage (Mussida & Sciulli, 2025). In Spain, the approval of the so-called "Rider Law" (Ley de Riders) in 2021, which mandates employment contracts for delivery riders working through digital platforms, represents an attempt to curb the spread of such precarious arrangements. However,

the enforcement and scope of this regulation remain limited to specific activities, and broader segments of the self-employed population continue to face high poverty risks.

Structural precariousness extends beyond the self-employment category and refers to the overall increase in unstable, low-quality jobs that lack sufficient security, predictability, and adequate income to prevent poverty. The Spanish labour market, characterised by high shares of temporary contracts and significant underemployment, exhibits features of what Standing (2011) has described as the "precariat": a class of workers with insecure and fragmented employment trajectories, limited career prospects, and inadequate social protection. In this sense, structural precariousness includes both wage earners and self-employed individuals whose employment conditions place them at the margins of economic stability.

Educational attainment and skill levels interact with these patterns of precariousness. Workers with lower levels of formal education are overrepresented among the self-employed poor and among those engaged in precarious wage employment (Eurofound, 2017; OECD, 2018). Moreover, sectoral segmentation reinforces these risks: employment in low-wage, low-productivity sectors not only increases individual exposure to poverty but also limits opportunities for upward mobility and earnings improvement over time (Gómez-Álvarez & Gómez, 2017).

Finally, collective bargaining coverage and employment regulation also influence the extent to which structural precariousness translates into in-work poverty. Spain's system of sectoral collective agreements provides some wage floors above the statutory minimum, but coverage remains weaker among self-employed workers and small firms, where precariousness is most prevalent (Malo, 2018). Recent labour market reforms have sought to restore the primacy of sectoral bargaining over firm-level agreements, with the potential to enhance wage setting and employment stability. However, significant challenges persist in extending these protections to non-standard forms of work.

In conclusion, vulnerable self-employment and structural precariousness constitute critical components of the labour market dynamics that sustain in-work poverty in Spain. These factors highlight the need to consider not only standard employment relationships but also the broader spectrum of atypical and marginal work arrangements in any analysis of working poverty.

### 2.2. Family structure, care responsibilities and their connection to in-work poverty

Household composition and family responsibilities are the second fundamental piece of the in-work poverty puzzle. In-work poverty has a dual nature: it depends not only on a person's employment and earnings, but also on their family circumstances (Peña-Casas & Latta, 2004; Ibáñez et al., 2024). A worker's income might be adequate to escape poverty if they are single or if their partner also earns, but the same income could leave them "working poor" if they are the sole provider for a family.

One of the clearest findings in the literature is that single-earner households face dramatically higher poverty risks than dual-earner households. Nolan and Marx (2014) emphasize that the majority of Europe's working poor are in households where only one adult is employed. In Spain, too, many working-poor individuals are sole breadwinners supporting dependents (often in low-paid jobs), whereas two-earner families – even if one or both earn low wages – are far less likely to be below the poverty line (Eurofound, 2017). This reflects the arithmetic of income pooling: a household with two earners has two income streams to bolster total resources, and typically benefits from economies of scale in consumption. By contrast, a lone worker carrying the economic responsibility for an entire household may struggle to generate sufficient income to meet the needs of all members. Family size and composition heavily influence the threshold at which poverty is measured (through equivalised income) and the actual living standards.

Households with children have higher required expenditures, and thus higher poverty thresholds, making it easier for a given income to fall short. Accordingly, having dependent children is associated with a greater risk of in-work poverty across Europe (Eurostat, 2020). In Spain, working households with multiple children, especially three or more, show substantially higher poverty rates than childless couples with similar earnings (INE, 2021). Large families often rely on a single income source or one and a half incomes (for instance, one full-time earner and one part-time earner), which may not stretch far enough once divided among many mouths. Single-parent households are particularly vulnerable.

A single parent who works – even full-time – has to provide for themselves and their children on just one salary, which often results in an income below the poverty line if that salary is modest. European data confirm that single parents in work have among the highest in-work poverty rates of any household type (Eurofound, 2017). In the Spanish context, Ibáñez et al. (2024) identify the presence of children and, especially, single parenthood as major contributors to what they term "severe in-work poverty" (where both individual earnings are low and household needs are high). Single mothers, for example, face a double challenge: they are often concentrated in lower-paid jobs (due in part to childcare-related career interruptions or part-time work) and they have no spousal income to supplement their earnings. As a result, even with

employment, a single mother's household may remain below the poverty threshold – a reality reflected in Spain's elevated poverty rates among single-adult-with-children families.

The distribution of work within households – often termed work intensity at the household level – is a key mediating factor. In-work poverty is strongly linked to households where the adults have a low work intensity (for instance, one worker in a household that potentially could have two, or one full-time equivalent shared between two adults). During the economic crisis, many Spanish households adjusted by having a second member (often a spouse) seek work to make ends meet, which helped some families avoid poverty. Conversely, in households where only one adult worked, poverty risks remained high or even worsened. Lanau & Lozano (2022) note that the stagnation of Spain's in-work poverty rate despite job growth can be partly explained by the composition of that job growth – many new jobs were taken by secondary earners in previously jobless households (improving those households' situation from unemployment to working-poor, thus appearing as a rise in working poverty), while many single-earner households continued to struggle. In essence, whether a working individual is poor depends greatly on who else is in the household and what they contribute. A low-paid worker living with affluent relatives may not be poor, whereas an average-paid worker supporting several dependents could be.

Gender roles and care responsibilities further connect family structure to in-work poverty. In many Spanish households, as in much of Europe, women more often assume unpaid care duties – looking after young children, elderly parents, or disabled family members. This unequal distribution of care can lead to women working fewer paid hours or interrupting their careers, thereby lowering household earned income. For example, a mother may switch to a part-time schedule or leave the workforce temporarily due to childcare constraints, leaving the family with effectively one full-time salary. If that salary is not high, the risk of poverty increases. Eurofound (2017) observed that households where one partner (often the woman) is out of the labour force due to care responsibilities have a higher incidence of in-work poverty, even if the other partner is employed, compared to dual-earner households with similar demographics. In Spain, where public childcare for children under 3 is limited and traditional gender norms persist in some regions, many women reduce their labour supply after having children (Gómez-Álvarez & Gómez, 2017). This "care penalty" in work participation contributes to poverty: families become single-earner or one-and-a-half-earner units, with constrained income. Additionally, when women do re-enter work, it is often in part-time or flexible jobs that accommodate care duties but tend to be lower-paid. The OECD (2021) notes that a high share of Spanish women in part-time work cite family reasons, and this group faces higher poverty rates than

women working full-time. In short, inadequate support for reconciling work and family life – such as insufficient childcare services or lack of flexible but decent-paying jobs – can translate into higher in-work poverty, especially for households with young children.

The extended family can sometimes mitigate in-work poverty in Spain, though this is a double-edged sword. It is common for young working adults with low incomes to live with their parents or for multi-generational households to pool resources. In some cases, this family solidarity prevents poverty – for instance, a low-earning youth living in their parents' home may not be counted as poor because the total household income (parents' pensions or wages plus the youth's income) is above the threshold. However, where the entire extended family has low resources (e.g. parents are also low earners or unemployed), cohabitation can mean multiple poor workers under one roof. Fundación FOESSA (2019) reports that many Spanish households in structural poverty include at least one working member, indicating that while family support networks are important, they cannot always compensate for insufficient labour income. Moreover, reliance on family support may mask individual poverty without truly lifting living standards – a young person's wage may be too low to form an independent household, effectively "hiding" potential poverty in a larger family unit.

Household factors also interact with the labour market characteristics of jobs. For instance, part-time work is less likely to lead to poverty if the worker's partner is employed, as noted. Conversely, temporary and unstable employment is particularly problematic when both partners in a couple have insecure jobs, or when a single parent cycles between low-paid jobs and unemployment. The Great Recession illustrated this in Spain: many dual-earner couples coped better if at least one partner kept a stable job, whereas families where both partners were in sectors hit by temporary layoffs or contract non-renewals often fell into poverty (Muñoz de Bustillo & Antón, 2018). The presence of social support (or lack thereof) for families – such as child benefits, discussed in the next section – also mediates how family structure translates to poverty outcomes. In summary, family structure and care responsibilities critically condition the link between employment and poverty. In-work poverty tends to concentrate in households with a single earner or many dependents, such as single-parent families and large families with one income (Eurofound, 2017; Ibáñez et al., 2024). On the other hand, dual-earner couples and smaller households have a much lower risk. The "individual" and "household" dimensions of in-work poverty are deeply intertwined (Raitano et al., 2019). This insight reinforces that policy responses must address not only job quality and wages but also family support systems. The next section will examine how public policies and the welfare state influence in-work poverty – either by exacerbating it (through gaps in the safety net or tax burdens on

low-income workers) or alleviating it (through redistributive transfers, tax credits, or wage regulations).

## 2.3. The role of public policies and welfare systems

Public policies and the design of the welfare system play a pivotal role in shaping in-work poverty outcomes. Even with the same labour market conditions and household structures, countries can have very different rates of in-work poverty depending on how much their tax-benefit systems support low-income working families (Peña-Casas & Latta, 2004; OECD, 2019). In broad terms, policies that "make work pay" – such as adequate minimum wages, earned income tax credits, or child benefits – can reduce in-work poverty by boosting net incomes. Conversely, the absence of supportive policies, or a tax system that heavily burdens low earners, can leave workers more exposed to poverty. Here we consider several key policy areas: wage regulation, taxation, direct income transfers, and social protection programs, with a focus on Spain's context.

### *2.3.1. Taxation, transfers and minimum wage policies*

Minimum wage policy is one of the most direct tools affecting in-work poverty. By setting a legal wage floor, the government can lift the lowest wages and reduce the incidence of very low pay. Spain's minimum wage (Salario Mínimo Interprofesional, SMI) has historically been relatively low as a proportion of the median wage – around 40%–50% of the median for much of the 2000s and 2010s – which limited its effectiveness in reducing working poverty. However, in 2019 and subsequent years, Spain enacted substantial minimum wage increases, with the monthly SMI rising from €736 (2018) to €900 (2019), and reaching €1,080 by 2023 (paid in 14 installments). These hikes were designed explicitly to combat in-work poverty and wage inequality. Early evidence indicates that they significantly increased earnings for workers at the bottom of the distribution (ILO, 2020). For example, many full-time workers previously earning around €700–800 per month were lifted closer to the poverty threshold for a single-person household. This likely helped some single workers escape poverty and provided additional income to poor households. Oxfam Intermón (2020) welcomed the policy, arguing that a higher wage floor is a necessary step to ensure "decent work" yields a living income.

That said, the minimum wage has limitations: it benefits only those employed; it does not directly help those without work or those who, due to part-time hours, still have low monthly earnings; and if set too low it has negligible impact on poverty, while if set very high it could, theoretically, affect employment opportunities for some groups (though evidence of significant job loss in Spain's case has been mixed). Overall, a

robust minimum wage is seen by most experts as a complementary but insufficient measure – it needs to be paired with other supports to fully address in-work poverty (Marx & Nolan, 2014).

Taxation affects the net income of workers and thus their poverty status. In Spain, low-wage workers benefit from income tax allowances that often exempt earnings up to a certain threshold, meaning many minimum-wage workers pay little or no income tax. This is positive for reducing poverty, as it leaves more take-home pay. However, Spain (like many European countries) relies heavily on regressive consumption taxes (VAT) which take a larger relative bite from low-income households' budgets. Moreover, social insurance contributions are levied from the first euro of earnings (for both employees and employers). These payroll taxes can cumulatively be substantial for low earners – for instance, a worker earning €1,000/month might contribute over €60 to social security, and their employer roughly €300, costs which indirectly can suppress take-home pay or employment. The OECD (2018) noted that the tax wedge (taxes and contributions minus cash benefits) on low earners in Spain is around the OECD average, but higher than in countries that offer targeted tax relief for working poor (such as earned income tax credits in the UK or US). Spain does have a small tax credit ("reducción por rendimientos del trabajo") for low-income workers, but its effect is modest. Essentially, Spain's tax system has not been highly effective in redistributing income toward low-income working families, especially when compared to countries that use the tax code to top up earnings of the poor. López-Laborda, Marín-González & Onrubia (2024) point out that the combination of direct and indirect taxes in Spain can leave working-poor households only slightly better off than they would be on gross earnings alone, since whatever small benefits they gain from low income tax rates can be offset by consumption taxes and lack of refundable credits.

Social transfers and in-work benefits are the other side of the equation – these include any cash benefits that supplement household income. In-work poverty can be mitigated if low-income workers receive adequate social support, such as child allowances, housing subsidies, or income-tested benefits. Spain's welfare system, traditionally, has been oriented more towards pensions and unemployment benefits (for those out of work) than towards supporting the working poor. Unlike countries such as France (with its Prime d'activité) or the UK (Working Tax Credit/Universal Credit), Spain until recently had no nationwide in-work benefit scheme to augment low wages.

The main forms of support relevant to low-income working families in Spain have been: means-tested regional minimum income schemes and the more recent national Ingreso Mínimo Vital (IMV) introduced in 2020. The IMV is a guaranteed minimum

income aimed at reducing extreme poverty; it can benefit some working poor households if their incomes are below the threshold and they meet other conditions. However, in practice, many working poor do not qualify (for instance, if they work enough hours to be above the very low cut-off, or if their household income slightly exceeds the IMV threshold which is around 50% of median income for a single person). Early evaluations of IMV showed relatively low take-up and a large exclusion of the "working poor" segment it might have helped (Ministerio de Inclusión, 2021).

Child benefits in Spain are minimal – a means-tested child allowance of around €30 per month per child for low-income households – much lower than the child benefits in most EU countries. Consequently, the overall redistributive impact of Spain's transfer system on in-work poverty has been limited.

Eurostat analysis suggests that Spain's at-risk-of-poverty rate for workers drops only a couple of percentage points after accounting for taxes and transfers, whereas in countries like Ireland or Finland, the drop is much larger due to more generous supports (Eurostat, 2020). OECD (2023) estimate that if Spain were to implement transfer policies on par with the EU average, its working poverty rate would be significantly reduced. Likewise, EAPN España (2022), the national anti-poverty network, has repeatedly highlighted that a considerable share of Spain's working poor receive little to no social assistance – they "fall through the cracks" because they are not unemployed (and thus not receiving unemployment benefits) but their earnings are still insufficient.

Social protection coverage for non-standard workers is another policy aspect relevant to in-work poverty. In Spain, standard employees have access to contributory benefits (like unemployment insurance and sick leave) which can prevent them from falling into poverty if they lose their job or become ill. However, temporary and part-time workers, as well as self-employed individuals, historically have lower coverage or entitlement. For example, temporary workers often do not accumulate enough contribution time to qualify for lengthy unemployment benefits, and the self-employed (autónomos) have until recently not been covered by unemployment insurance at all. This means that these workers are more vulnerable to income shocks and may remain in low-paid, precarious work out of fear of no safety net. Recent reforms have aimed to extend protections – e.g. the "Tarifa Plana" and other schemes for self-employed, and improved access to benefits for part-timers – but gaps remain. The ILO has stressed that strengthening social protection, including for those in atypical employment, is crucial to tackle in-work poverty, because it reduces the risk that workers will fall into poverty when work is disrupted and can empower workers to seek better jobs without fearing total destitution (ILO, 2016). In Spain's case, during the COVID-19 pandemic, the government implemented ERTE furlough

schemes that subsidised wages for workers whose jobs were suspended. This emergency measure prevented a potential surge in in-work poverty by ensuring that millions of workers continued to receive around 70% of their wages while they were temporarily idle, rather than dropping to zero income (Mussida & Sciulli, 2025). This highlighted how active policy intervention can shield working families from poverty during crises.

Apart from cash transfers, public services can indirectly affect in-work poverty by reducing necessary expenses. Affordable childcare, for instance, enables parents (especially mothers) to engage in paid work; without it, families might reduce their employment (and income) or spend a large share of earnings on private childcare, both scenarios that can lead to poverty. Spain has made progress in preschool provision for 3-5 year olds (universal access), which has helped many mothers join the workforce. However, for children under 3, coverage is uneven and often expensive, which can trap some parents in low work intensity and consequently low income (OECD, 2021).

Similarly, housing costs play a role: Spain has relatively limited social housing or housing allowance programs for low-income workers, so a low-paid family in an expensive city like Barcelona might be pushed into poverty primarily due to rent costs. While housing policy is outside the direct scope of labour income, it is part of the welfare system that determines disposable income left for a family.

Labour market policies and institutions also merit mention. Active labour market policies (ALMPs) such as job training, re-skilling programs, or employment services can help low-income workers transition to better-paying jobs, thus reducing in-work poverty in the long run. Countries that invest heavily in ALMPs (e.g. Denmark, Germany) often see shorter durations of low-paid work and better upward mobility. Spain has traditionally underinvested in ALMPs relative to the EU average (Eurofound, 2017), which may contribute to fewer opportunities for the working poor to upgrade their skills or find higher-paying employment.

Another institution is collective bargaining: where unions and employer associations negotiate industry wages, they can compress the lower end of the wage distribution above poverty levels. Spain's collective bargaining system covers a large portion of workers via sectoral agreements, and recent legal changes (repeal of some 2012 labour reforms) aim to strengthen collective bargaining. If successful, this could raise wage floors in sectors that are not covered by the statutory minimum wage alone (for example, sectors where agreements have traditionally set starting wages 10-20% above SMI). Stronger bargaining outcomes for low-wage sectors would complement the minimum wage in fighting in-work poverty (OECD, 2019). On the flip side, if

bargaining erodes or if contracts are not enforced (say, in informal work situations), then many workers can effectively earn below the negotiated minima, thus remaining poor.

In Spain, one institutional peculiarity is the high share of informal or unregistered work. Workers in the informal economy often have very low earnings and no access to social security, which virtually guarantees poverty conditions. Tackling in-work poverty thus also involves measures to formalise work and extend legal protections to all workers. Spain's inspections and regularisation efforts (including recent steps to bring platform workers under formal employment contracts via the "Rider Law" of 2021) are policy efforts aimed at this issue. By ensuring that more workers are legally employed with at least minimum wage and contributions, these policies seek to reduce the incidence of working poverty rooted in informality.

In conclusion, public policies can significantly alleviate or exacerbate in-work poverty. Redistributive policies – whether through wage-setting (like minimum wage laws and collective bargaining) or through the tax-benefit system (like tax credits, child benefits, and income support) – have been shown to reduce the depth and incidence of in-work poverty in countries that employ them effectively (Eurofound, 2017; Marx & Nolan, 2014). Spain's relatively high in-work poverty rate in the 2010s can be partly attributed to policy gaps: a low wage floor (until recent increases), limited in-work transfers, and social protection schemes that did not fully reach or benefit low-income workers (Lanau & Lozano, 2024; EAPN, 2020). Recent improvements, such as the rising minimum wage and the creation of the IMV, indicate a policy recognition of the problem. Still, as Ibáñez et al. (2024) argue, a comprehensive strategy is needed – combining decent work conditions with supportive family policies and targeted transfers – to effectively lower in-work poverty. Having examined labour market and household factors alongside policy influences, we can now summarize the key determinants of in-work poverty identified across the literature.

## 2.4. Summary of key determinants according to academic literature

The literature converges on a small set of factors that consistently explain why some workers, despite being employed, still fall below the poverty threshold.

- **Employment and earnings capacity**:
    - Low hourly pay in lower-value-added sectors limits disposable income.
    - Too few working hours or involuntary part-time schedules suppress total earnings.

  - Job instability—temporary contracts, intermittent spells of self-employment—adds income volatility and weakens bargaining power, curbing career progression.

- **Household structure and needs**:
  - Single-earner families, particularly those with children or dependent relatives, face greater pressure because one pay-cheque must stretch further.
  - Lone parents or large households amplify this strain, while young or low-skilled earners often lack the resources to cushion shortfalls.
- **Welfare-state and labour-market institutions**:
  - Adequate minimum wages and broad collective-bargaining coverage can lift pay floors.
  - Well-designed tax credits, child benefits and social assistance offset low market income; gaps leave households exposed.
  - Affordable childcare, activation policies and the effective inclusion of non-standard workers in social insurance help second earners join the labour force and stabilise household resources.
- **Macro-economic and structural context**:
  - Recessions, regional disparities and long-run shifts—technological change, globalisation, sectoral transformation—shape the availability and quality of jobs.
  - Where productivity gains fail to translate into broad wage growth, pockets of low pay persist and in-work poverty endures.

In short, when limited earnings capacity meets high household needs and public transfers fall short, the risk of remaining poor while employed rises sharply.

# 3. Data and methodology

## 3.1. Rationale for the focus on Spain and on recent data

Spain provides a compelling case study for in-work poverty due to its persistently high incidence of this phenomenon in a European context. In fact, Spain has one of the highest in-work poverty rates in Europe. Recent studies note that about 16% of working households in Spain were living in poverty (as of late 2010s), a level markedly above the EU average. This risk is concentrated in certain household types – for example, over one-third of Spanish households with dependent children and only one employed member were in poverty. Such figures underscore the severity of the issue

and the importance of understanding its determinants in the Spanish context. Focusing on Spain allows us to examine in-work poverty in a labour market characterised by high rates of temporary employment, pronounced wage inequality, and strong family-based social patterns, all of which may shape the working poor population differently than in other countries. Emphasis is placed on using the most recent available data to ensure the analysis reflects the current socio-economic reality. The study utilises data from 2024 (specifically, incomes from 2023 – see Section 3.2), capturing the post-COVID recovery period and recent policy interventions. In Spain's case, the late 2010s and early 2020s saw significant changes such as substantial minimum wage increases and labour market reforms aimed at reducing precarious employment. By analysing the Encuesta de Condiciones de Vida 2024 (ECV 2024) – the latest wave of Spain's survey on income and living conditions – we can assess in-work poverty after these developments. Using up-to-date data enhances the policy relevance of the findings, as trends in employment and poverty may have shifted in response to economic growth and new social policies. In summary, focusing on Spain with recent data provides a timely and context-rich understanding of in-work poverty in a country where this issue is acute and undergoing change.

### 3.2. Data Source: EU-SILC and its Spanish Implementation (ECV 2024)

The analysis is based on microdata from the European Union Statistics on Income and Living Conditions (EU-SILC), specifically its Spanish implementation. EU-SILC is a harmonised annual survey conducted across EU Member States to collect information on household income, poverty, social exclusion, and living conditions. It is the reference source for comparative statistics on income distribution and poverty in Europe, ensuring that data from Spain are comparable with that of other countries. In Spain, the survey is known as the Encuesta de Condiciones de Vida (ECV) or Living Conditions Survey (LCS), carried out by the Spanish National Statistics Institute (INE). The ECV 2024 is the edition used in this study, which was collected in 2024 and captures income data for the calendar year 2023. Using this survey wave means our income measurements correspond to 2023 (the reference year for earnings and poverty status), while demographic and employment information refer to the situation at the time of interview in 2024. This timing is advantageous for analyzing in-work poverty in the aftermath of recent economic conditions.

The ECV follows a complex survey design to ensure representativeness of the Spanish population. It uses a stratified two-stage sampling, with census tracts as first-stage units and households as second-stage units. The survey is also a rotating panel, wherein each sampled household is interviewed for up to four consecutive years. For

the 2024 wave, the effective sample comprises roughly 72,000 individuals living in approximately 25,000 households across Spain. This large sample provides sufficient cases of the working poor to enable detailed analysis. Data collection in 2024 was conducted via a combination of online, telephone, and face-to-face interviews, adhering to Eurostat's quality standards. The resulting microdata are organised into separate files: a household-level file (with information on total household income, housing conditions, etc.) and an individual-level file (with personal data such as age, education, labour status, and individual income components). For this study, these files were merged using household identifiers to create a unified analytic dataset. Each record in the final dataset corresponds to an individual respondent, augmented with attributes of their household. This structure allows us to analyze individual outcomes (such as a person's poverty status) in conjunction with both personal characteristics and household-level factors (e.g. total household income, number of household members). Prior to analysis, we performed data cleaning and transformation steps (detailed in Section 3.4) to construct the specific variables required for our models, such as equivalised income and indicators of work intensity. The survey weights provided by INE/Eurostat were retained and later used in all analyses to adjust for the sampling design and to ensure that results are representative of the Spanish population (see Section 3.5). Overall, the EU-SILC/ECV 2024 microdata offer a rich and reliable source of information for examining in-work poverty in Spain, with comprehensive coverage of incomes and living conditions and variables that align with standard definitions of poverty and labour market status.

### 3.3. Operational definition of in-work poverty: inclusion criteria and unit of analysis

In this study, in-work poverty (IWP) is defined in line with the standard Eurostat indicator of the "in-work at-risk-of-poverty" rate. An individual is considered to be in-work poor if two conditions are simultaneously met: (1) the person is in work (i.e. gainfully employed according to specified criteria), and (2) the person's household is at risk of poverty based on disposable income. Following the EU convention, the poverty condition is determined by a relative poverty threshold – specifically, having an equivalised disposable household income below 60% of the national median. Disposable household income refers to total income from all sources (after taxes and including social transfers) received by the household over the reference year. Equivalised income is calculated to account for household size and composition, using the modified OECD equivalence scale (which gives a weight of 1.0 to the first adult, 0.5 to each additional adult, and 0.3 to each child). This yields a per-person income measure that is comparable across households of different sizes. For 2023 (the income reference year), the 60% median poverty line is computed from the ECV

data itself, and individuals are classified as "poor" if their household's adjusted income falls below that cutoff. The dependent variable in our analysis is a binary indicator for in-work poverty status: it is set to 1 for individuals who are working and simultaneously in a poor household, and 0 for individuals who are working but live in a non-poor household. (By construction, those not in work are not part of the analytic sample for IWP, as explained below.)

The definition of "in work" for this purpose requires clarification of the inclusion criteria and unit of analysis. We restrict our analysis to working-age individuals who are substantially attached to the labour market. In practice, we include individuals aged roughly 18 to 64 who were employed for a significant portion of the reference period. Eurostat's methodology for the in-work poverty rate typically considers persons who worked at least seven months out of the year as "in work". Adhering to this approach, we include respondents who reported being employed or self-employed for at least half of 2023 (≥7 months). This criterion excludes very short work stints, ensuring that the focus is on people with meaningful labour market participation over the year. Both employees and self-employed workers are encompassed by this definition of in-work status (reflecting the EU indicator's coverage of anyone "at work (employed or self-employed)". We also exclude individuals above retirement age (65+) from the analysis, since regular employment beyond that age is less common and their income dynamics often involve pensions rather than earnings. The unit of analysis is the individual worker. Each person meeting the above employment criteria is treated as an observation in the dataset, with their poverty status determined by their household's income. It is important to note the interplay of individual and household levels in this operational definition: poverty is assessed at the household level (as resources are typically shared within households), but we attribute that status to the individual if they are an active worker in that household. For example, if a respondent worked 8 months in 2023 and her household income is below the threshold, she is classified as in-work poor. If her household income is above the threshold, she is a non-poor worker. By defining the outcome this way, the analysis solely focus on the working poor population and enables identification of factors distinguishing poor workers from non-poor workers. This approach is consistent with the literature on in-work poverty, which treats it as a phenomenon affecting individuals but fundamentally measured via household income (Marx & Nolan, 2012; Halleröd et al., 2015). All analyses in subsequent sections will therefore pertain to the sub-sample of employed individuals (meeting the criteria) and use the binary IWP indicator as the key outcome of interest.

### 3.4. Dependent and independent variables: construction and justification

Dependent Variable: The dependent variable is the In-Work Poverty (IWP) status, a dichotomous variable. It equals 1 if an individual is in-work and in a poor household (i.e. equivalised disposable income < 60% median), and 0 if the individual is in-work and not in poverty. Using a binary outcome allows for straightforward interpretation in the logistic regression models (odds of being working poor versus not).

Independent Variables: A range of independent variables were derived from the survey data to serve as predictors of in-work poverty. These were chosen based on theoretical considerations from prior research and on the need to capture the multiple dimensions that can influence a working person's poverty status. We group these explanatory variables into three broad categories – (a) individual-level demographic characteristics, (b) household-level composition and work intensity measures, and (c) employment-related factors – and detail their construction and rationale below.

- **Individual Demographic Characteristics:**
    - **Age**: Measured in years (as a continuous variable). We include age to account for life-cycle effects on income and poverty risk. Younger workers may have lower earnings or more precarious jobs, while older workers (up to retirement age) might have more stable careers or dual-earner households. If non-linear effects are expected, age can be tested in quadratic form, but primarily it serves as a control for experience/seniority.
    - **Gender**: A dummy variable for gender (Female = 1, Male = 0) is included. Gender is relevant given observed disparities in the labour market – for instance, women often face lower wages and a higher incidence of part-time work, yet some studies find that in-work poverty is strongly linked to household composition, which can mask individual wage disparities. We include gender to observe any differences in the likelihood of being working poor, controlling for other factors. This can shed light on whether female workers in Spain face higher in-work poverty, potentially reflecting issues like the gender pay gap or the prevalence of women as second earners.
    - **Educational Attainment**: Education is coded into three levels: low (up to lower secondary), medium (upper secondary or post-secondary non-tertiary), and high (tertiary education). This categorisation follows standard ISCED groupings. Education level is a critical determinant of employment quality and earning capacity; workers with low education tend to have lower-paying, less secure jobs and thus are more vulnerable to poverty. Including education allows us to control for

human capital – we expect higher education to be associated with lower odds of in-work poverty, as found in numerous studies of poverty risk.

- **Other personal characteristics** such as migration background or region can be included. For instance, whether the individual is an immigrant or foreign-born might influence in-work poverty due to integration and job sector differences. Likewise, regional controls (e.g., dummies for Spain's autonomous communities or an urban/rural indicator) could account for cost-of-living and labour market variation.

- **Household Composition and Work Intensity:**
  - **Household Type**: We constructed categorical variables to represent the composition of the individual's household, distinguishing types such as single adult, single parent with children, couple without children, couple with dependent children, and other household (e.g., extended family). This was derived from the household roster in the ECV, using the number of adults and dependents. The justification for including household type lies in evidence that in-work poverty is strongly mediated by family structure.
  - **Number of Earners**: Closely related to household type, we include a variable for the number of employed persons in the household. This captures the degree of income pooling from multiple workers. Rather than use the raw count (which correlates with household size), we operationalised this as an indicator of whether the individual is the sole earner in their household or not. Being the only worker (single-earner household) is expected to elevate in-work poverty risk, as there is no second income to buffer against low earnings. Conversely, a household with two or more earners has a better chance of staying above the poverty line, even if one income is modest. Prior research has highlighted that the presence of a second earner is often more crucial for avoiding in-work poverty than the wage level of the first earner. Including this variable tests that insight in the Spanish data. (This concept will also be indirectly captured by the low work intensity measure below, but we include both for thoroughness.)
  - **Household Work Intensity** (Low Work Intensity): We calculated each household's work intensity – the ratio of the total number of months that working-age adults in the household worked during the year to the total number of months they theoretically could have worked. Following the EU's definition of "very low work intensity," we then created a dummy variable low_work_intensity (lowWI) which equals 1 if the household work intensity is 20% or less, and 0 otherwise. In other words, lowWI = 1 typically indicates that household adults

worked only a small fraction of the year (e.g., a couple where only one person worked a few months, or a single adult who worked intermittently). This indicator aligns with one of the components of the AROPE (At Risk of Poverty or Social Exclusion) metric, capturing households with marginal labour force attachment. We include low work intensity as an independent variable because it encapsulates underemployment at the household level – an important predictor of poverty. We expect individuals living in very low work intensity households to have significantly higher odds of being poor, since limited overall employment translates to low cumulative earnings. However, note that our analytic sample already requires the individual to have worked ≥7 months, so truly "very low work intensity" households are largely filtered out by design (such households often have no one working a substantial part of the year). Even so, this variable can identify cases where, for example, the person in question worked part-year but other adults in the home did not work at all, resulting in low overall work input. The inclusion of lowWI is justified by the idea that household labour supply, not just individual effort, matters for escaping poverty. This helps differentiate whether an individual's poverty is due to their own low earnings or due to limited work contribution from other household members.

- **Employment-Related Factors**: These variables characterize the individual's employment situation and job attributes, which are directly relevant to their earning level and poverty risk. In-work poverty often reflects not just whether one works, but what kind of work one has – stable or unstable, well-paid or low-paid, full-time or part-time, etc. To capture these aspects, we include:
    - **Employment Type (Occupational Status)**: We distinguish between employees (wage and salary workers) and self-employed individuals. A binary variable is used (Self-employed = 1, Employee = 0) to mark those who are self-employed (including freelancers, business owners, and unpaid family workers). This distinction is important because self-employment in Spain ranges from high-earning professionals to precarious own-account workers. Notably, EU statistics often find higher poverty rates among self-employed workers compared to employees, due to volatile incomes and less access to social protection. By controlling for employment type, we can see if being self-employed carries an extra risk of in-work poverty, all else equal. We anticipate that self-employed workers may have higher odds of being working poor, reflecting either genuinely low incomes for some or under-reporting of income.

- **Contract Type (Job Stability)**: For those who are employees, we include an indicator for temporary employment (Temporary contract = 1, Permanent contract = 0). Spain's labour market has historically been characterised by a high prevalence of temporary jobs, which generally offer lower pay and less security. We thus expect that holding a temporary contract is associated with a greater likelihood of in-work poverty. To incorporate this in the model, we created a dummy based on the individual's main job contract status as reported in the survey. If the individual is self-employed, this variable is not applicable; effectively, the baseline group in our regression will be permanent employees, with controls for temporary employee and for self-employed. This way, we compare poverty outcomes across three groups: permanent workers (most stable), temporary workers (potentially more precarious), and the self-employed. The inclusion of contract type is justified by the link between job instability and low earnings– temporary jobs in Spain often come with fewer hours or benefits and lower cumulative annual income, increasing poverty risk.
- **Working Time (Full-time vs Part-time)**: We include a variable for part-time employment (Part-time = 1, Full-time = 0) in the individual's main job. Part-time workers earn less on an annual basis by virtue of fewer hours, and if the part-time status is involuntary (common in some service jobs or among those with care responsibilities), it may lead to poverty even if hourly wages are decent. Using the self-reported usual hours, we classify anyone working less than full-time hours as part-time. We anticipate that part-time workers have higher odds of in-work poverty than similar full-time workers, unless they have other household income to compensate. This variable helps capture the effect of under-employment or time-related underinvestment in work on poverty outcomes.
- **Low Wage Indicator**: To directly assess the impact of low pay, we constructed a low wage indicator. Drawing on the concept of low-wage employment used in EU analyses, we define low wage as earning two-thirds or less of the median gross wage. In practice, using the microdata, we approximated this as follows: for employees, we computed their gross hourly wage (using information on annual earnings from work and hours or months worked), then determined the threshold corresponding to 2/3 of the national median hourly wage. A dummy variable LowWage = 1 indicates those employees whose wage falls below this threshold. (For self-employed individuals, wage data is not directly comparable, so the LowWage flag is primarily

meaningful for employees; self-employed were given LowWage = 0 by default in analysis, and we rely on the separate self-employed indicator to capture their earnings dynamics.) This low wage variable is included to test one of the intuitive drivers of in-work poverty: simply not earning enough per hour or per job to lift a household out of poverty. We expect a positive association between being low-paid and being in poverty, but it may not be deterministic – for instance, a low-wage worker with a second earner at home might not be poor. By including LowWage alongside household composition variables, we can see the net effect of low pay on poverty. This is also in line with debates in the literature about the extent to which in-work poverty is a wage problem versus a household employment problem. If low wages remain a significant predictor even after accounting for household factors, that underscores the need to address wage levels; if not, it suggests that the number of earners and other factors matter more.

- **Occupation and Sector**: In addition to the above, we control for the broad occupational group and/or sector of employment of the individual. Different occupations and industries have different wage distributions (for example, workers in agriculture or certain services might earn less than those in finance or public sector). Using the ISCO occupational classification provided in ECV, we include dummy variables for major occupational categories (e.g., managers/professionals, clerical, services, skilled trades, elementary occupations, etc.), with one category as the reference. Similarly, we include sector controls (perhaps distinguishing agriculture, industry, construction, services, or public vs private sector employment). These variables are included mainly as control variables to soak up structural wage differentials that are not the main focus of the study. By controlling for occupation and sector, we ensure that our key coefficients (e.g., for contract type or education) are not biased by the fact that certain types of workers cluster in low-paying industries. We do not delve into the detailed effects of each occupation/sector in our interpretation, but their inclusion improves the model's specification.

Each independent variable above was carefully constructed from the survey data (e.g., using specific questionnaire items or derived variables in the ECV microdata) and checked for consistency. Where necessary, recoding was done – for instance, educational categories were harmonised, and outliers in wage calculations were reviewed. All monetary variables (income, wages) are expressed in annual 2023 euros.

## 3.5. Analytical strategy

Our analytical approach proceeds in three main stages. First (Section 3.5.1), we conduct a descriptive and correlational analysis to understand the data and simple relationships between each factor and in-work poverty. Second (Section 3.5.2), we employ an exploratory feature selection technique, the Cross Entropy (CE) algorithm, as a complementary tool to identify which combinations of variables have the strongest association with in-work poverty. This serves as a data-driven check on our model specification. Third (Section 3.5.3), guided by the insights from theory, descriptives, and the CE exploration, we specify and estimate a multivariate logistic regression model for in-work poverty. Throughout these stages, we use appropriate weighting and statistical methods suited to the complex survey data. The overall goal is to ensure that the final regression model is well-grounded (both theoretically and empirically) and that we have addressed potential issues like multicollinearity and model uncertainty in the process.

### *3.5.1. Descriptive and correlational analysis*

We begin by examining the data through descriptive statistics and simple correlations, which provides intuition and validation for the more complex analysis to follow. All descriptive calculations use the survey weights to reflect population estimates. In practice, we apply the appropriate weight for individual-level statistics (e.g., the percentage of workers in poverty, mean age of workers) and purely household-level aggregates. By doing so, we correct for any sampling bias and ensure that, for example, the overall in-work poverty rate we compute matches official figures within sampling error.

The descriptive analysis has several components:

- Overall incidence of in-work poverty and subgroup breakdowns: We calculate the weighted in-work poverty rate among workers in Spain and then break it down by key demographic groups. For instance, we compute the IWP rate separately for men and women, for different age brackets, for native vs immigrant workers, etc. This involves using weighted proportions and can be presented in tables or charts.
- Distribution of income and wages: We examine the distribution of equivalised household income for our working sample, comparing those in poverty vs. not in poverty. This could be done through density plots or cumulative distribution functions to illustrate how the incomes of the working poor cluster below the poverty line while others are more spread above it. Similarly, we look at the distribution of individual wages and the incidence of low pay.

For example, we might plot the wage distribution and mark the low-wage threshold (60% median) to see what fraction of workers earn near or below that level. These distributions can reveal, for instance, whether there is a significant wage gap between those who are poor and those who are not (we expect the working poor to have lower median wages). We will also compute the proportion of workers who are low-wage and what percentage of those are in poor households, versus the proportion of working poor who are low-wage. This informs how tightly low earnings and poverty overlap.

- Cross-tabulations and bivariate relationships: We produce cross-tabulations (contingency tables) and charts to explore how in-work poverty relates to categorical variables one at a time. For example, we tabulate IWP status by contract type (permanent vs temporary) to see the difference in poverty rates, or by full-time vs part-time status, or by education level. Each of these is effectively a bivariate analysis, usually expressed as a percentage of the subgroup that is in poverty (with weighting). These comparisons provide initial evidence of which factors might be important determinants. However, at this stage, we refrain from causal interpretation; these are raw associations. We do, however, note where multiple disadvantages coincide.

Through the descriptive analysis, we expect to paint a picture of the profile of the working poor in Spain. This includes identifying which categories (e.g., low-educated, temporary, part-time, single earners with children, etc.) have higher incidence of in-work poverty. While these are unconditional statistics, they provide valuable context and justification for the multivariate analysis: they suggest which variables should be in the model and hint at what relationships we might find after controlling for everything. Notably, if some factor shows little difference in poverty rate in descriptives, it might have a weaker role (though it could still matter once others are controlled). On the other hand, if a factor shows a large disparity (say, self-employed vs employees), we will verify if that persists in the regression.

### *3.5.2. Exploratory analysis using Cross Entropy as a complementary tool for variable selection*

In addition to the theory-driven approach, we employ a Cross Entropy (CE) algorithm as an exploratory tool to assist in variable selection for our model. The Cross Entropy method is a stochastic optimization technique originally developed for estimating rare event probabilities (Rubinstein, 1997) and later adapted to solve combinatorial optimization problems. In our context, the combinatorial problem is to find the optimal subset of independent variables that best predicts the outcome

(in-work poverty). We treat this as a feature selection problem: among the many candidate variables described in Section 3.4, which combination yields the strongest predictive performance for distinguishing between IWP = 1 and IWP = 0 cases?

Rationale for using Cross Entropy: Traditional stepwise regression or manual trial-and-error could be used to select variables, but those approaches risk overlooking complex interactions among predictors or getting stuck in local optima (e.g. adding one variable at a time might miss combinations that are only jointly important). The CE method offers a more global search of the model space by considering many subsets of variables and iteratively honing in on the best one. It has been shown in other applications (e.g., credit risk modeling) to outperform some greedy algorithms, providing greater flexibility in variable selection and sometimes achieving better model accuracy. Using CE is an addition that complements the theoretical selection – it helps ensure we did not bias the model by only our expectations and can reveal non-obvious predictors that improve fit.

How the Cross Entropy algorithm works for feature selection: The CE algorithm can be summarized in iterative steps:

- Initialization: We represent any subset of predictors as a binary string (vector) where each element corresponds to a variable (1 = included, 0 = excluded). Initially, we assign equal probability (e.g., 0.5) for each variable being included, reflecting no prior knowledge of which are important.
- Random sampling: Using these inclusion probabilities, the algorithm randomly generates a large number (N = 125) of candidate variable subsets (binary vectors). This means each subset is like a possible model specification.
- Evaluation: For each candidate subset, we fit a predictive model and evaluate its performance using a chosen metric. In classification problems, one might use misclassification rate or AUC; in our case we could use the log-likelihood (deviance) or an error metric on predicting IWP. Essentially, we need a scalar score of model quality for each subset.
- Elite selection: We then select the top-performing subsets – the best n models with the lowest error or highest likelihood (this is the "elite" set). We keep the best 25 models out of 125. These elite solutions represent the most promising combinations of variables in the current iteration.
- Probability update: Next, we update the inclusion probability for each variable based on the elite set. Concretely, for each variable, we look at how many of the elite models included that variable (i.e. the proportion of elite solutions where the variable's bit = 1). We then set the new probability for that variable to this proportion. In effect, variables frequently present in the best models

get a higher inclusion probability going into the next iteration, whereas those seldom present get a lower probability.

- New iteration: Using the updated probabilities, we generate a new sample of model subsets and repeat the evaluation, elite selection, and update steps. Over successive iterations, the probability vector converges towards 0 or 1 values, indicating a near-deterministic choice of including or excluding each variable as the algorithm homes in on an optimal or near-optimal subset
- Stopping criterion: We iterate until a maximum number of iterations is reached (in this case, 25) or until the changes in probabilities become very small (convergence). At that point, we have an estimated optimal subset of predictors.

The output of the CE algorithm for our purposes is twofold. Firstly, it identifies a particular combination of variables that optimizes the model's fit (the best subset found). Secondly, it provides an estimate of the importance of each variable, which can be interpreted as the probability of inclusion in the optimal solution. In fact, after final convergence, each variable has an inclusion probability (either close to 0, close to 1, or intermediate if the algorithm hadn't fully converged). We can treat those probabilities as a ranking of variable importance: a higher probability means the variable was consistently useful in the best models. In our adaptation, we take the final inclusion probabilities from the CE procedure in the Random Forest methodology as indicative of each variable's predictive power for IWP. For example, if "temporary contract" ends up with a probability of 0.95, it suggests that almost all of the top-performing subsets included this variable, hinting it is very important; whereas if "gender" has a probability of 0.2, it was rarely included in good models, implying a weaker predictive role.

We implement the CE algorithm on our dataset as a complementary exploratory step. It is important to emphasize that this is not the primary method for inference – rather, it is used to guide and confirm our model specification:

- We run CE on a portion of the full sample (via cross-validation) to identify variables and then validate the final model on the same data.
- The performance metric chosen is failure rate minimization, which aligns with logistic regression objectives.
- We include nearly all candidate variables in the CE search initially (perhaps excluding any that are very collinear or virtually constants).
- After running CE, we examine which variables have high inclusion probabilities. We expect, based on theory, that variables like low wage, household work intensity, contract type, etc., will emerge as important – and

this would validate our theoretical choices. If CE suggests additional variables or interactions that we hadn't considered, we take note.

- We do not blindly accept the CE-selected model; instead, we compare it to our theory-driven model. In practice, there should be substantial overlap.

By using CE, we add a data-driven check against omitted variable bias and model misspecification. For instance, if household ownership status turned out not to improve prediction, CE would down-weight it, and we might omit it from the final model for simplicity. Conversely, if an unexpected variable (say, the sector of employment or a specific type of benefit) has strong predictive power, we ensure it's accounted for. The CE method is essentially performing a form of robust feature selection under uncertainty, leveraging random sampling to explore many combinations that would be infeasible to test exhaustively.

We aim for an outcome where the CE algorithm helps refine our choice of variables for the logistic regression, yielding a model that is both parsimonious and high-performing. Additionally, as an academic contribution, this showcases a machine learning approach in a socio-economic analysis, highlighting the interdisciplinary methodology.

It should be noted that the CE analysis is exploratory; our inference about which factors "matter" will ultimately rely on the logistic regression results (with significance tests, etc.). The CE step is not used to produce final estimates or p-values – it is used to select or prioritize variables. Therefore, it does not violate any assumptions of the regression (since it's separate from it), though one must be careful about overfitting when using such techniques. The computational complexity of CE is manageable here because the number of potential variables is on the order of a few dozen, and logistic models are quick to estimate.

In summary, the Cross Entropy method provides a systematic way to probe the data for the most relevant predictors of in-work poverty, complementing our theoretical framework. It serves as a robustness check on our model specification. We will report on which variables were reinforced by the CE feature selection in Chapter 4 (Results), discussing any cases where the algorithm's suggestions led us to adjust the model (e.g., adding or removing a control). This approach strengthens the credibility of our methodology by reducing subjectivity in variable selection.

### *3.5.3. Specification of the logistic regression model based on theoretical considerations, descriptive evidence and Cross Entropy results*

The guiding principle was to balance theoretical coherence with data-driven parsimony. Economic theory and prior European evidence (Nolan & Marx, 2014; Eurofound, 2017) indicate that in-work poverty is shaped by three concentric layers:

- Individual endowments – human capital, health and migrant status;
- Job quality – pay level, contract security, weekly hours and employment status;
- Household context – work-intensity, dependency ratio, housing costs, and the buffering role of public transfers.

A strict theory-only model (as the one presented in Table 3.1), however, risks overlooking empirical nuances, while a purely algorithmic approach can discard variables that are substantively important but only weakly correlated within one sample. We adopted a hybrid workflow.

- Step 1 (exploratory) – All cleaned predictors were fed to a Cross-Entropy search meta-heuristic that iteratively fits Random-Forest classifiers and retains top-performing variables. Twenty-four variables survived this screen, including low-wage status, work-intensity (WI), single-parent flag and tenure. The CE-RF model (Section 4.2.5) achieves the highest pseudo-$R^2$ (0.221) but omits education and gender—factors central to labour-economics theory.
- Step 2 (confirmatory) – Guided by the CE-RF shortlist and by theoretical salience, we built four cumulative logit models. Each layer adds a thematically coherent block, permitting intuitive comparison of marginal effects:

*Table 3.1. Theory-based logistic regression stepwise scheme summary*

| Model | Block added | Rationale | Indicators retained |
|---|---|---|---|
| **M1** | Demographics | Baseline control for labour-supply capacity and discrimination | sex, age dummies (captured by age≤40 HH flag), education, chronic illness, migrant status |
| **M2** | Labour-market quality | Wage, intensity and contract mechanisms | low-wage dummy, full-time/part-time, temporary contract, self-employment, job tenure |
| **M3** | Household structure | Dependency ratio and intra-household labour sharing | WI index, presence of children, single-parent, home-ownership, HH age composition, main-earner gender & education, family-type dummies |

| | | | |
|---|---|---|---|
| **M4** | Policy buffers | Redistribution / insurance & unobserved regional costs | share of means-tested transfers (P_Transf), pension share (P_Pens) |

*Source: Own elaboration*

Variables flagged by CE-RF but absent from the theoretical catalogue (e.g. specific "multi-unit dwelling" proxy) were included only if they remained significant ($p < 0.10$) after the full block was introduced. Conversely, a priori key variables (education, gender) were kept even if their significance waned, because they facilitate interpretation and interaction testing.

Estimation details used are as follows:

- Sample – 30 729 employed persons aged 18-64, residing in private households, with complete labour, income and household rosters in the 2024 ECV.
- Outcome – $IWP_i = 1$ if the equivalised disposable household income is below 60 % of the national median and the person worked ≥ 7 months in the reference year.
- Estimator – Maximum-likelihood logit with household-clustered robust SEs; person weights re-scaled to preserve original weight sum.
- Baseline categories – Male, primary schooling or less, native-born, healthy, employee, permanent-contract, part-time, wage ≥ 60 % median hourly rate, childless dual-adult household, tenant/other tenure, WI = 1, no transfers.
- Goodness-of-fit metrics – Pseudo-$R^2$ (McFadden), log-likelihood, and out-of-sample AUC from 10-fold cross-validation.

The stepwise theory-guided model was retained as the primary explanatory tool because:

- It preserves variables essential for policy interpretation that CE-RF discarded for statistical parsimony (education, gender, full-time hours).
- It aligns neatly with the literature's causal ordering, facilitating comparison with earlier Spanish studies (Tejero, 2017; Ibáñez et al., 2024).
- Its performance (pseudo-$R^2$ = 0.175, AUC = 0.789) is only modestly below the CE-RF's, while offering clearer pathways for simulation of policy levers (e.g. raising WI from 0.5 to 0.75, or increasing child benefit generosity).
- The CE-RF model is retained as a robustness benchmark; its superior fit confirms that the dominant drivers isolated by the cumulative theory model—low pay, low WI, single-parenthood, many children, non-EU origin and lack

of housing assets—are indeed those with the greatest predictive power in the data.

Together, the dual-track strategy satisfies both the econometric requirement for model adequacy and the substantive requirement for policy-relevant storytelling.

### 3.6. Limitations of the Methodological Approach

While the data and methods chosen for this study are rigorous and appropriate for the research question, it is important to acknowledge their limitations. These limitations temper the conclusions and suggest caution in interpretation, as well as opportunities for further research to address them:

- Cross-sectional design: The analysis is based on a single year of data (ECV 2024, incomes for 2023) and is therefore cross-sectional. This means we observe correlations between variables and in-work poverty status at one point in time, but we cannot firmly establish causal relationships or the direction of effects. For example, we find an association between receiving certain social transfers and in-work poverty, but it may be that being poor leads to receiving transfers (not vice versa). Similarly, the cross-sectional snapshot cannot capture dynamics such as movements into or out of in-work poverty over time. Longitudinal data (tracking individuals across years) would be needed to study causal pathways and sequences (e.g., a job loss causing entry into in-work poverty, or a wage increase causing exit), but that is outside the scope of this undergraduate dissertation. Thus, our findings should be interpreted as indicative of risk factors and correlates, not definitive proof of causation.
- Limited time frame – recent but singular: Although using the latest data is a strength for relevance, it also means our results are contingent on the context of 2023. That year might have particularities (for instance, unusual inflation or specific policy effects from the recent minimum wage hike) that are not generalizable to other periods. Without multiple years to compare, we cannot tell if the relationships we find are stable over time or if they might be transient. For example, if 2023 had a post-COVID employment recovery that temporarily altered the profile of working poor, our results capture that mix. In-work poverty rates can change with economic cycles; a boom or a recession could change the composition of who is working and poor. Therefore, one should be careful in extrapolating our quantitative estimates beyond the immediate period studied.
- Survey data and measurement error: The ECV survey data, while high quality, is subject to some measurement issues. Income data in surveys can suffer from under-reporting or reporting error, especially for self-employment income or

informal work. If some incomes are under-reported, some individuals might be misclassified as poor or non-poor. We rely on INE's data processing, which mitigates error (and uses administrative data when possible for accuracy), but some error remains. Additionally, the hours and months worked are self-reported and might be approximate for some respondents (though likely reasonably accurate in aggregate). Any measurement error in key variables (e.g., if some people misreported working 7 months when it was 6 or vice versa) could lead to slight misclassification in our IW indicator or other covariates, potentially biasing results toward null (classical measurement error generally attenuates associations). We assume these errors are not systematic enough to overturn findings, but it's a limitation to acknowledge.

- Definition of "in-work" and exclusion of marginal employment: Our criterion of ≥7 months worked to be considered "in-work" means we exclude individuals who had very limited employment in the year (even if they are poor). While this aligns with official definitions, it does mean our study does not cover some forms of precarious work, such as people who had a couple of short gig jobs in the year and remained in poverty. Those individuals are arguably experiencing a form of in-work poverty too, but are statistically categorized as unemployed or inactive poor. Thus, our findings apply to those with relatively substantial work involvement. If one's interest is in the broader concept of "the working poor" including marginally attached workers, our analysis might understate the extent of the problem. However, focusing on ≥7 months ensures we analyze situations where work is a primary status, which is conceptually clearer. This limitation is essentially a scope choice: it might omit some edge cases but improves comparability and clarity.
- Omitted variables and model specification: Despite our comprehensive set of independent variables, there are factors we could not include due to data unavailability or to maintain focus. For instance, regional cost-of-living differences within Spain are not explicitly modeled. The poverty threshold is national, but living costs differ (e.g., living in Madrid or Barcelona is more expensive than in a rural area). Two households with the same income might have different real standards of living depending on region, potentially affecting the in-work poverty experience. We did not include region fixed effects in the regression, partly because that would absorb much variation (since poverty lines are national, region dummies would mainly capture cost differences and some labour market differences). Excluding region means our model might have residual spatial autocorrelation – if certain regions have both higher poverty and particular characteristics (say, higher temporary employment rates), our coefficients might partly capture regional effects.

Future research could examine results by region or include region-specific poverty thresholds.

Similarly, occupational level or job skill is not directly included (we included sector and education, but not a measure of occupation like professional vs unskilled). Occupation correlates with education and sector, but a direct measure (e.g., ISCO job category) might explain additional variance in wages. We opted not to overload the model with every possible variable to keep it interpretable, so some less critical variables were left out. This means there is a chance of omitted variable bias if an excluded factor is correlated with both included factors and the outcome. The risk is mitigated by our inclusion of the major known determinants, and our use of CE helped check if any important variable was missing. Nonetheless, no model can include everything; for example, personality traits or negotiation skills that affect wages are unobserved, and any effect they have might be subsumed in, say, the education coefficient as a proxy.

- Multicollinearity and interpretability: Some of our independent variables are interrelated (as discussed in model design). Although we addressed multicollinearity to some extent (and will verify VIFs), it can still impair the interpretation of individual coefficients. For example, education and low wage are connected: including both means the "education" coefficient is the effect of education holding wage level constant, which is a bit abstract (since education mainly works through wage). If high multicollinearity were present, standard errors would inflate and make it hard to declare significance. We believe the model is structured to minimize this (education picks up broader skill effect, low wage picks up the actual earning outcome, etc.), but readers should be aware that regression coefficients in a model with many controls are conditional effects. They should be interpreted with the phrase "ceteris paribus" – sometimes that means imagining a scenario that is rare in reality (e.g., a high-educated person earning a low wage, if both education and low_wage are in the model). We have taken care to ensure such scenarios exist in data or that variables remain meaningful when combined. Still, the complexity of the model could make some findings less straightforward. We mitigate this by focusing on key results and by possibly providing marginal effects for interpretation (average marginal effects can be easier to digest than raw log-odds).
- Cross Entropy feature selection limitations: While the CE algorithm is a powerful tool, it is not without drawbacks. It involves random sampling and has hyperparameters (like number of iterations, sample size, elite fraction). If set poorly, CE might converge to a suboptimal subset or overfit the noise in the data. We mitigated this by reasonable choices and by focusing on

consistent findings rather than any one run of the algorithm. Additionally, using CE on the same data that we then use for final estimation could, in theory, introduce a bit of overfitting – we effectively "peeked" at which variables work best. However, since we have a large sample and ultimately include many variables anyway (not just a razor-thin selection), the risk is low. Still, it's a limitation that our feature selection was data-driven; in a confirmatory analysis one might prefer pre-specification of the model purely on theoretical grounds or test the CE-selected model on a holdout sample. Due to scope and data constraints, we did not implement a separate validation sample for CE; we rely on the signs and significance in the final model to judge the relevance of variables.

- Survey scope and population: The ECV survey covers private households and thus excludes people in institutions or homeless, who are not part of the sampling frame. By definition, our study doesn't consider the working poor who might not have stable housing (though many homeless are not working, some could be employed but without a home – a scenario of extreme in-work poverty we can't capture here). This is a standard limitation of household surveys, but worth noting as it means the most vulnerable cases might be missing. Additionally, certain groups like seasonal migrant workers might be underrepresented if they don't participate in the survey or don't have a registered address. Our weighted results assume representativeness, but any hard-to-reach population could be undercovered, potentially biasing findings if those groups have different poverty dynamics.
- Interpretation of policy variables: Some independent variables we include (like receipt of benefits, or being in certain contract types) are directly influenced by policy and institutional context. While we can identify associations (e.g., single parents have higher odds of poverty), attributing causality or evaluating policy effectiveness is beyond what our method can do robustly. For instance, if we find that having a higher share of income from social transfers correlates with higher poverty (which it likely will, mechanically), this should not be interpreted as "social transfers cause poverty". It's rather that those households are poor hence eligible for transfers. Our methodology is not a program evaluation; it doesn't use instrumental variables or longitudinal approaches that would be needed to isolate policy impacts. We mention this to caution against any simplistic reading of coefficients on such variables – they are included to control for differences among households, not to conclude on the efficacy of welfare programs. In the limitations, we therefore clarify that endogeneity is a concern for variables like benefits shares or even work intensity (household work intensity can be low partly as a result of poverty if, say, poor health caused both no work and poverty).

- Parsimonious vs comprehensive tension: We aimed for a comprehensive model, but there's always a trade-off between including more variables (to avoid omitted bias) and keeping the model parsimonious (to avoid overfitting and to maintain degrees of freedom, though the latter is not a big issue with our sample size). If anything, our model errs on the side of including a broad array of controls. One limitation is that this can make the model a bit complex to communicate. However, since this is a final-year undergraduate project, clarity is important. We will address this by focusing on the most impactful variables in the discussion and perhaps grouping variables when presenting results (e.g., "job-quality variables such as low wages, temporary contracts, and part-time status all show strong effects" rather than drowning in each of the many dummies). The methodological point is that adding many covariates could introduce noise or multicollinearity as mentioned, but also it could reduce the interpretability of the model as a whole. We acknowledge that a more parsimonious model might be easier to interpret, but we opted for a fuller model to ensure robustness.
- External validity: Our findings are specific to Spain. While that is intentional, it means the results may not directly apply to other countries with different labour market institutions or social welfare systems. In-work poverty is known to vary widely across Europe – our model's implications (e.g., the large role of temporary contracts) might be very context-specific. This is not so much a limitation of the method as a scope limitation of the study. A comparative approach could enrich the conclusions but was beyond our scope.

In conclusion, we recognize that no single-method study can be without weaknesses. We cannot claim to identify causal effects, but we can identify strong associations consistent with theoretical expectations. We cannot cover every possible determinant, but we cover the major ones documented in literature. Being transparent about these limitations ensures that our conclusions – for example, the identification of key risk factors for in-work poverty in Spain – are properly qualified and understood as part of a complex socio-economic reality. Future research (as discussed in the conclusion chapter) could build on this by using longitudinal data, exploring policy experiments, or comparing multiple countries to address some of these limitations. For the purposes of this dissertation, our methodological approach remains solid for an explanatory analysis of the characteristics of in-work poverty in Spain, within the acknowledged bounds of what it can and cannot reveal.

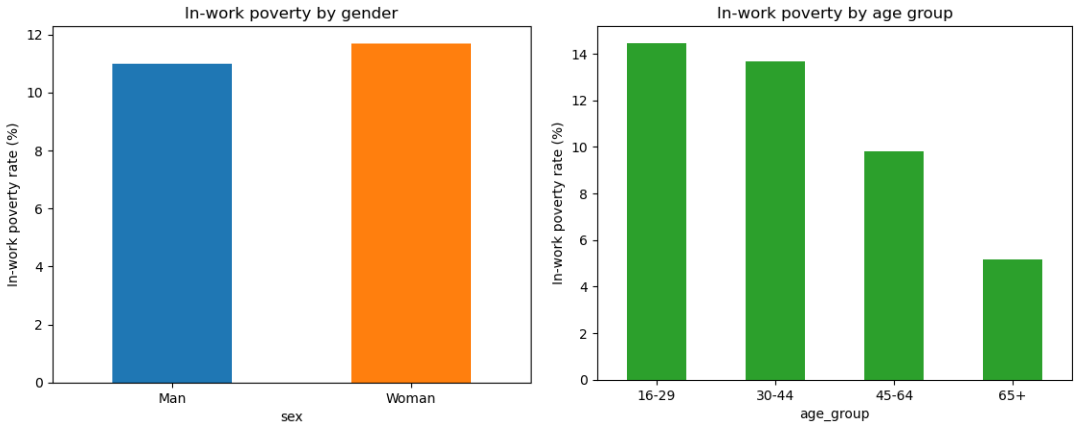
# 4. Empirical results

## 4.1. Descriptive overview of in-work poverty in Spain

### *4.1.1. Overall incidence and recent trends*

The overall incidence of in-work poverty in Spain is high. According to the latest data, about 11.3% of employed persons live in households below the poverty threshold (60% of median income) despite being in work. This corresponds to roughly 2.35 million working poor out of ~20.9 million workers, underlining that more than one in ten workers is affected. By historical standards, the in-work poverty rate remains elevated, though it has declined slightly from mid-2010s peaks. For instance, around 2014–2017 the rate hovered in the 12–13% range before edging down to the current ~11% level as the labour market recovered. Even so, in-work poverty rate continues to exceed the EU-27 average (circa 9–10% in recent years) and is among the highest in Western Europe. This persistent prevalence suggests that economic growth alone has not fully resolved the problem of in-work poverty in Spain.

*Figure 4.1. In-work poverty rate by demographic characteristics*




*Source: Own elaboration*

In terms of basic demographics, the risk of in-work poverty is fairly similar for men and women (see Figure 4.1) . Women have a slightly higher incidence (11.7%) than men (11.0%), but the gender gap is small. This near-parity reflects the broad reach of in-work poverty across the workforce, although women's marginally higher rate may relate to their higher likelihood of part-time employment and lower average earnings. There is a much sharper disparity by age (see Figure 4.1). Younger workers face markedly higher in-work poverty rates: about 14.5% of workers aged 16–29 are poor despite working, compared to 13.7% for those 30–44. Poverty risk then drops significantly for older workers, falling to 9.8% for ages 45–64, and only ~5% for those 65+ (a small group of late-career or post-retirement workers). In other words, in-work poverty decreases with age. Younger workers are the most vulnerable – likely due to entry-level low wages, unstable early-career jobs and internships, and shorter job tenure – whereas prime-age and older workers tend to have more stable

employment and higher earnings, reducing their poverty risk. The very low rate among working seniors may partly reflect a selection effect (only those with sufficient income stay in or re-enter work at 65+) and the role of pensions or accumulated assets. Overall, the profile of in-work poverty in Spain is broad: it affects both men and women almost equally, but is disproportionately concentrated among younger workers. The following sections examine how job characteristics and household circumstances further shape these poverty risks.

### *4.1.2. Distribution by labour market characteristics*

The risk of in-work poverty in Spain varies markedly across different labour market groups. Contract type is one important dimension (see Figure 4.2): workers on temporary contracts face a higher in-work poverty rate (approximately 13.2%) than those on permanent contracts (around 11.0%). This gap suggests that contract instability contributes to poverty risk, albeit the difference is moderate. Notably, low pay is more prevalent among temporary workers. About 21.6% of temporary employees are low-wage, compared to 17.5% of permanent-contract workers who earn low wages. This indicates that temporary jobs not only carry slightly elevated poverty rates but are also more likely to be low-paid. However, because permanent contracts make up the majority of employment, the bulk of Spain's low-wage workers are nevertheless found in permanent positions. In fact, nearly 90% of all low-wage workers have a permanent contract (versus only ~12% on temporaries), reflecting the overall composition of the workforce. This composition effect explains why many of the working poor are in ostensibly stable jobs – permanent employees – even though the incidence of low pay is somewhat higher under temporary contracts.

Working hours show an even larger disparity in in-work poverty outcomes. Part-time workers experience an in-work poverty rate of 18.4%, roughly double that of full-time workers (about 9.6%). This substantial gap aligns with expectations that reduced working hours (and thus earnings) elevate poverty risk. It is interesting, however, that fewer part-timers are low-paid (only 12.4% earn low wages) compared to full-time workers (19.1%). This somewhat counter-intuitive finding likely arises because the definition of "low-wage" here is relative to hourly earnings – many part-time workers may earn close to median hourly wages (for example, in public sector or skilled jobs) even if their total income is lower due to shorter hours. By contrast, a segment of full-time workers earn persistently low hourly wages, pushing nearly one in five full-timers into the "low-wage" category. Nevertheless, the higher poverty rate among part-timers underscores that working fewer hours (even at reasonable hourly pay) often results in insufficient household income. Indeed, focusing on employees alone, those in part-time employment have an in-work poverty incidence of roughly 17%, over

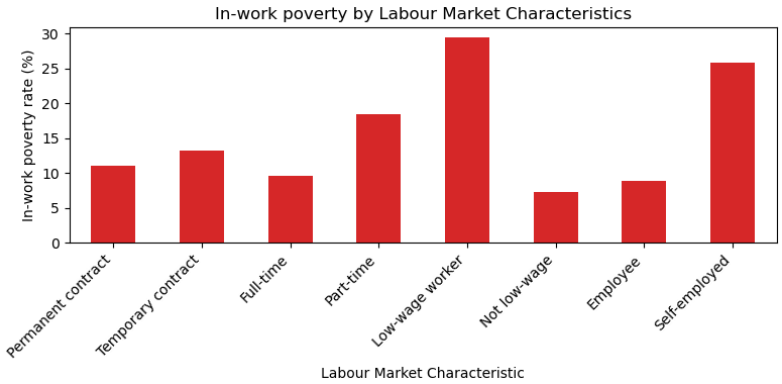
twice the rate of their full-time employee counterparts (around 7%). This reinforces that limited work hours, often involuntary or in low-paid sectors, significantly heighten the likelihood of poverty in work.

Unsurprisingly, low pay status itself is a strong predictor of in-work poverty. The analysis shows that 29.4% of low-wage workers in Spain are in poverty, a dramatically higher rate than the 7.3% observed among workers earning above the low-wage threshold. This nearly fourfold difference confirms that low hourly earnings often translate into inadequate monthly incomes at the household level. In-work poverty among the "not low-wage" group is relatively low (around 7%), reflecting that having at least median-level wages generally provides sufficient earnings to avoid poverty – especially if coupled with full-time hours or additional household income. By contrast, almost one in three low-wage workers are poor despite being employed. These figures highlight low pay as a key mechanism driving in-work poverty. It is also noteworthy that low-wage status cuts across various job types: a large share of low-wage workers are in what might be considered standard employment situations. For instance, over 87% of low-wage individuals work full-time, and around 88% hold permanent contracts. Low pay is therefore not confined to atypical or part-time jobs – many full-time, permanent workers still earn below the low-wage threshold. This underlines that while certain job characteristics (temporary contracts, part-time hours) accentuate the risk, the condition of low pay per se is pervasive and places workers at high risk of poverty regardless of their contract or hours.

Finally, employment status (employee vs self-employed) reveals the widest gulf in poverty rates. Only about 9.0% of employees live in in-work poverty, compared to a striking 25.8% of self-employed workers. Thus, self-employed individuals are nearly three times as likely as employees to be working poor. This disparity may reflect lower earnings among self-employed (including many own-account workers and freelancers), as well as potentially volatile incomes and weaker access to social benefits. The prevalence of low pay among the self-employed is extremely high: roughly 76% of self-employed workers are low-wage earners, versus just 8.8% of employees. In other words, three out of every four self-employed persons in Spain earn below the low-wage threshold – an astonishingly large proportion – whereas fewer than one in ten employees do so. This stark contrast helps explain the high poverty incidence in self-employment. Many self-employed Spaniards are essentially working poor by virtue of very low earnings. It also means that the self-employed constitute a disproportionate share of all low-wage workers. Indeed, despite forming a minority of the labour force, self-employed individuals account for about 57% of all low-wage workers (with employees comprising the remaining 43%). This

imbalance suggests that policies addressing in-work poverty cannot overlook the self-employed, as they form the core of the low-paid, high-poverty workforce in Spain.

*Figure 4.2. In-work poverty rate by labour market characteristics*




*Source: Own elaboration*

When labour market disadvantages combine, the in-work poverty risk escalates further, underscoring the compounding effect of multiple vulnerabilities. For example, a worker who simultaneously has a temporary contract and low wages faces an in-work poverty rate of roughly 28–29% – comparable to the very high poverty levels seen among low-wage permanent workers (~30%). In other words, whether a low-paid worker's contract is temporary or permanent, around one in three are poor, highlighting that low pay on its own is a critical determinant of in-work poverty. However, other overlapping disadvantages do make a notable difference. A particularly vulnerable profile is the part-time self-employed. Combining self-employment (with its high incidence of low earnings) and part-time work (fewer hours of income generation) results in an exceptionally high poverty rate. Estimates suggest that nearly 40% of self-employed people working part-time are in poverty – far above the overall self-employed average of 25.8%. Even self-employed individuals who work full-time have a high poverty incidence (~24–25%), but it surges for those who cannot secure full-time activity. Similarly, part-time work combined with a temporary contract yields a heightened risk: we find that over 20% of temporary part-time workers are in poverty, roughly double the rate of those in permanent full-time jobs (around 10%). In contrast, the most "secure" combination – a permanent, full-time job with decent pay – corresponds to the lowest in-work poverty likelihood. For instance, an employee who is full-time, on a permanent contract, and not low-paid has a very low probability of being poor (in many cases approaching the negligible levels). These cross-category results illustrate that each additional disadvantage – be it job insecurity, limited working hours, or low pay – adds to the poverty risk. Workers facing multiple disadvantages (for example, a temporary, part-time, low-paid self-

employed person) experience a cumulative penalty, placing them at the highest end of the in-work poverty spectrum.

In summary, the distribution of in-work poverty by labour market characteristics reveals a clear pattern: precarious employment conditions and low earnings greatly heighten the risk of poverty. Temporary contracts and part-time hours each on their own are associated with higher poverty rates, and being self-employed is associated with the highest risk of all. Crucially, low-wage employment emerges as a common thread linking these disparities – it dramatically elevates poverty risk across all categories. Furthermore, when these disadvantageous factors intersect in the same job (for instance, low pay coupled with insecure contracts or reduced hours), they reinforce each other's effects, leading to especially high incidences of in-work poverty. These findings underscore the importance of considering multiple job-quality dimensions together: tackling in-work poverty in Spain will require not only improving wages, but also addressing the instability of contracts and the insufficiency of working hours that leave many workers – particularly the self-employed and those in atypical jobs – vulnerable to poverty despite being employed.

### *4.1.3. Distribution by household structure*

The household composition of workers in Spain plays a crucial role in their likelihood of experiencing in-work poverty. The majority of Spanish workers live in relatively small households of one or two people – roughly 51% are in two-person households and about 32% in single-person households, according to the data. Larger households with three or more members are much less common (together accounting for under 17% of workers), and very large households of five or more are extremely rare (around 1% of workers). This distribution is important context because the incidence of in-work poverty differs markedly across these household types. In general, smaller households without additional earners tend to face higher poverty risk, whereas multi-person households often benefit from having multiple incomes or shared resources that reduce that risk.

Household poverty rates by size confirm this pattern. Workers living alone have an in-work poverty rate near 20%, among the highest for any household type. In contrast, those in two-person households have a much lower poverty rate (about 7.8%), likely reflecting the support of a partner's income or the pooling of two earnings. Three-person and four-person households – typically two adults with children – show even lower poverty rates (around 5%), suggesting that dual-earner couples with dependents are comparatively less vulnerable. Not a single case of in-work poverty was observed among workers in households of five or more members in the data (0% incidence), although this may be due to the very small sample of such

households. It does hint that in large family or extended-family households, multiple earners and shared expenses can virtually eliminate the risk of poverty for those who are working. Overall, these figures underscore that having more adult members (and potential earners) in the household strongly correlates with lower in-work poverty.

Beyond household size, certain specific household structures stand out for their high poverty prevalence. Single-parent households – though accounting for only about 1.4% of workers in the sample – experience by far one of the highest in-work poverty rates, at approximately 30%. This means nearly one in three workers who are lone parents (or the sole earners in single-adult families) live below the poverty line. The vulnerability is even more pronounced when the main income earner (MIE) in a single-parent family is a woman. In female-headed single-parent households, the in-work poverty rate reaches above 30%, a notably high level. This reflects compounding challenges: many single parents are women, and they often must provide for children on a single salary, without support from a partner's earnings. The data confirm that over half of all single-parent working households have a female main earner, and this subgroup faces the most severe poverty risk. By contrast, other household types with two adults (whether with or without children) have significantly lower poverty incidence, highlighting the importance of a second contributor to household income.

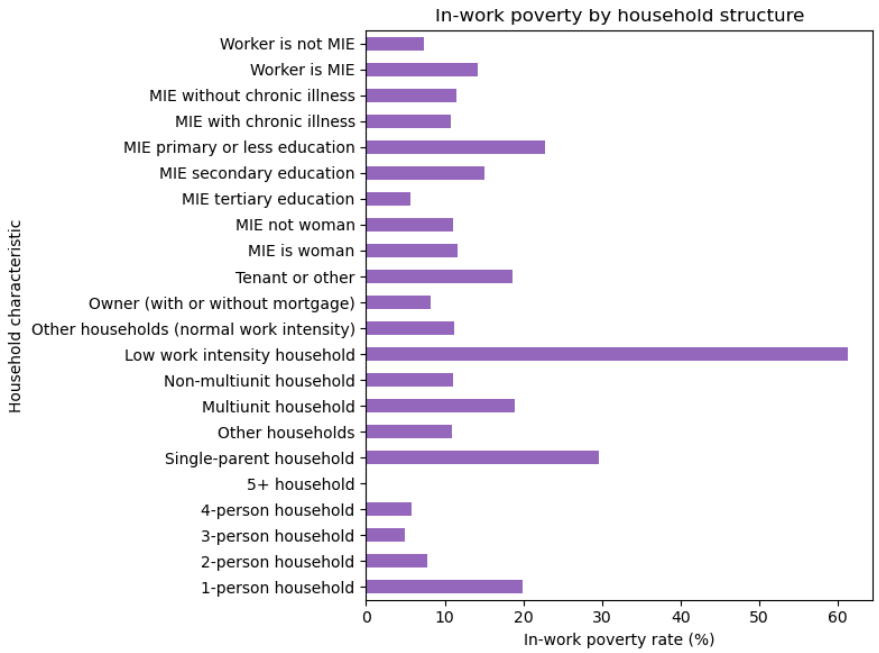
Another category with elevated poverty is multiunit households, where multiple family units share the same dwelling (for example, extended families or separate economic units living together). These households represent roughly 2.7% of workers – a small share – but have an in-work poverty rate of about 19%, substantially higher than the 11% seen in more typical single-unit households. This suggests that when families double up or live with relatives (often an indicator of economic necessity), their combined incomes still often fall short of the poverty threshold. In other words, the very fact that a household is "multiunit" may indicate financial strain that necessitates shared housing, reflected in the higher poverty rate. However, the multiunit poverty rate is not uniform across all cases; it is strongly influenced by the main earner's characteristics. For instance, if the main income earner in a multiunit household has tertiary education, the poverty rate in that group is only around 6% – comparable to the low poverty levels of tertiary-educated households in general. This implies that higher education (and presumably better-paying employment) can offset many disadvantages associated with complex household arrangements. On the other hand, multiunit households in which the main earner has low qualifications or a health limitation remain very vulnerable. Indeed, when a multiunit household's MIE has a chronic illness, the poverty rate is about 20%, essentially as high as the overall multiunit average. These interactions suggest that extended-family living, by itself, does not guarantee protection from poverty – it often goes hand-in-hand with other

risk factors, unless counterbalanced by a well-educated and presumably higher-earning primary breadwinner.

One of the most dramatic findings is the impact of household work intensity. Workers in low work intensity households (defined by Eurostat as households where working-age adults worked only a small fraction of the year) are extremely likely to be poor. The in-work poverty rate in such households is approximately 61%, by far the highest among all categories examined. In practical terms, this means a clear majority of workers in low-intensity households cannot earn enough to lift their families out of poverty. In fact, cross-tabulations reveal that certain subgroup of this category approach a 100% poverty rate. For example, in every observed case where a household was low-intensity and the main earner was a woman, the worker ended up in poverty. Similarly, low-intensity households led by a chronically ill main earner showed a 100% in-work poverty incidence. These extreme values underscore that insufficient labor participation within a household virtually guarantees poverty status, which is unsurprising since low work intensity often implies only one part-time job or intermittent work supporting the entire household. By contrast, households with normal work intensity (i.e. where adults maintain regular employment) have an in-work poverty rate around 11.2%, close to the overall average. This stark gap (61% vs 11%) highlights work intensity as a key driver of in-work poverty: when not enough work is being done relative to the household's needs, even those who are employed are overwhelmingly likely to be poor.

Housing tenure is another structural factor linked to in-work poverty outcomes. The data show that workers in tenant households (renters or those in free/subsidised housing) have a significantly higher poverty rate than those in owner-occupied homes. About 18.7% of workers living in rented or non-owned accommodation are in poverty, compared to only 8.3% of those in owner households. This nearly two-fold difference suggests that homeownership tends to coincide with greater financial stability among working households. Owners might have accumulated assets or enjoy lower housing costs (especially if mortgages are paid off), whereas renters often have less wealth and face ongoing rent expenditures, leaving them more vulnerable to poverty despite working. It is also possible that higher-income families self-select into homeownership, so tenure status here is partly capturing underlying income and wealth disparities. In any case, the tenure gap in poverty rates reinforces the idea that lacking property assets (and having to pay rent) is a significant disadvantage for in-work poverty risk.

*Figure 4.3. In-work poverty rate by household characteristics*




*Source: Own elaboration*

Finally, examining the main income earner's personal characteristics provides further insight into household poverty dynamics. The education level of the MIE emerges as a crucial protective factor: households whose main earner has tertiary education experience only about a 5.6% in-work poverty rate. This is the lowest of any subgroup and reflects the higher earnings and better job stability associated with higher education. By contrast, when the main earner's highest education is secondary school, the poverty rate jumps to 15.1%, and it surges to roughly 22.8% if the MIE has only primary education or less. Thus, each step down in educational attainment of the household's primary breadwinner corresponds to a markedly higher likelihood of the household being working poor. The gender of the main earner shows a more modest effect on poverty in aggregate, but still a discernible one. Households led by a woman have an in-work poverty rate of about 11.7%, slightly above the 11.0% for those led by a man. This gap is relatively small overall, suggesting that simply having a female main earner is not by itself a large risk factor once other variables are accounted for. However, as noted earlier, many female-led households are single parents or single-earner couples, which are contexts with inherently higher poverty risks. In fact, the cross-tabulations indicate that over 54% of single-parent working households have a female main earner, and within that group poverty is especially high (exceeding 30%). Thus, the slight overall gender gap in MIE poverty rates likely arises from these specific high-risk family situations rather than from gender per se. Meanwhile, the presence of a chronic illness or disability in the main earner does not show the expected disadvantage in the overall statistics – somewhat counterintuitively, households with an ill or disabled MIE have a poverty rate of 10.9%, which is

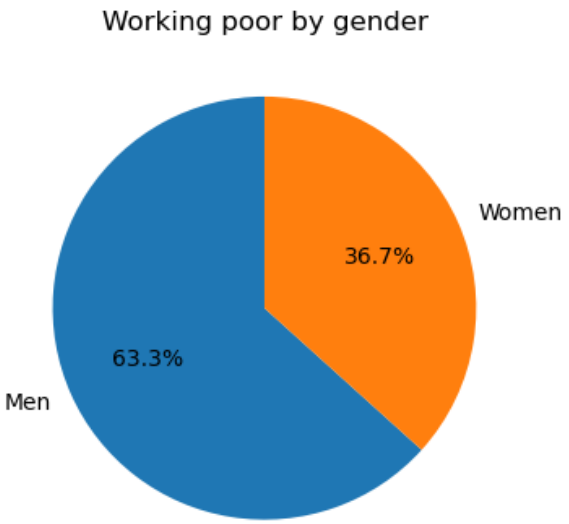
marginally lower than the 11.4% for households where the main earner has no chronic health issues. This difference is very small, implying that, on average, having a chronically ill main earner does not significantly change the poverty risk among those who are employed. One possible explanation is that individuals with serious health conditions who are able to remain in work might benefit from social transfers (disability payments, for example) or may be a self-selected group with relatively stable jobs, thereby offsetting what we might assume to be a vulnerability. In any case, the data does not show a large gap here, indicating that illness alone is not a strong predictor of in-work poverty once the person is employed.

Importantly, whether the worker themselves is the main income earner of the household or not makes a substantial difference in poverty outcomes. If the person in work (the survey respondent) is also their household's primary breadwinner, the in-work poverty rate is about 14.2%. In contrast, if the worker is a secondary earner (meaning someone else in the household earns more and is considered the MIE), the poverty rate drops to just 7.3%. In other words, a worker who is not the chief earner in their family is only about half as likely to be poor as a worker who is the sole or main earner. This makes intuitive sense: secondary earners typically reside in households where a principal earner provides a major share of income, which often keeps the household out of poverty. Many secondary earners are spouses or adult children contributing a second income, so even if their own wage is low, the presence of another full income in the household elevates total resources above the poverty line. Conversely, workers who are the main (or only) earners carry the entire responsibility of lifting their household's income, and if their wages are low or if there are multiple dependents, they are much more likely to fall short. Thus, the gap between 14.2% and 7.3% vividly demonstrates the protective effect of having multiple earners in the household. It also ties back to the earlier point about household size and composition: single-earner households (whether single adults or single-income families) are considerably more vulnerable to in-work poverty than those with two earners. In summary, the analysis by household structure reveals that in-work poverty in Spain is not just an individual phenomenon but closely linked to family circumstances. Smaller and single-parent households, low work intensity, renting tenure, and having only one income source all correspond to significantly higher poverty rates, while households with more earners, higher education, and homeownership enjoy much lower risks. These findings reinforce the idea that policies aimed at reducing in-work poverty must consider the household context – supporting second earners, childcare for single parents, encouraging full work participation, and similar measures – rather than focusing solely on individual wages.

#### *4.1.4. Socio-demographic profile of the working poor*

Having examined the incidence of in-work poverty by various factors, we now turn to the socio-demographic profile of those who are "working poor." In other words, among the population of employed people in poverty, what are their characteristics? Understanding the composition of the working poor can shed light on which groups are over-represented and the structural drivers behind in-work poverty.

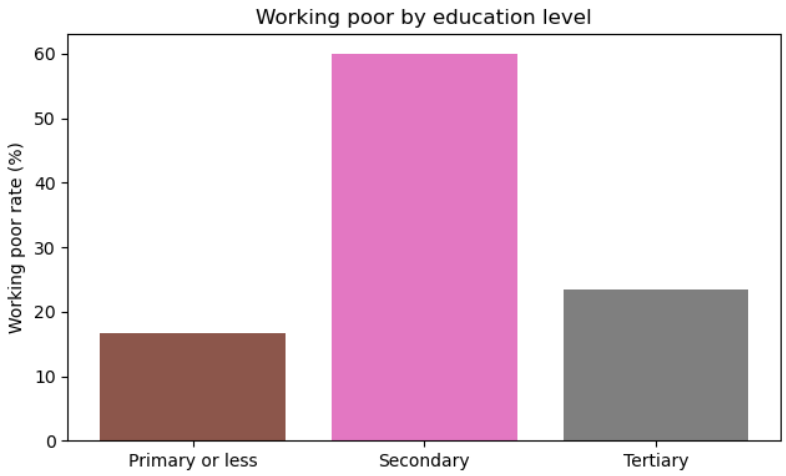
*Figure 4.4. Gender profile of the working poor*




*Source: Own elaboration*

One striking aspect is the gender composition of the working poor. Although men and women in Spain have nearly equal in-work poverty rates, men form a significantly larger share of the working poor in absolute terms. Specifically, about 63% of working poor individuals are men, and 37% are women. This imbalance reflects the underlying employment demographics – men constitute a slightly higher proportion of the labour force and work more hours on average (e.g. men are more likely to be full-time, whereas many women work part-time). As a result, even with similar poverty rates, men end up comprising roughly two-thirds of the working poor. It is notable that women's share (36.7%) is lower than men's despite women's slightly higher individual risk, indicating that women's overall labor force participation (especially in full-time jobs) is lower. In essence, the typical working poor person in Spain is more likely to be male, which is an important nuance given that poverty in the general population often disproportionately affects women. Here, because we focus on workers, the gender pattern aligns more with workforce participation. This does not diminish the hardships faced by the 37% who are women – rather, it highlights that policies addressing in-work poverty must address both men and women, with perhaps a focus on enabling more women to access stable full-time employment to reduce their poverty incidence.

*Figure 4.5. Education profile of the working poor*




*Source: Own elaboration*

In terms of age, the bulk of working poor are in the middle of their working lives. Contrary to what one might expect given the very high poverty rates among young workers, relatively few of the working poor are young. Only about 9.5% of working poor are aged 16–29. In contrast, the largest group is workers aged 45–64, who make up roughly 47% of working poor, and those 30–44 account for about 41%. A very small fraction (around 2%) are aged 65+. This age distribution might seem counter-intuitive at first, since younger workers have higher poverty rates individually. However, it reflects the age structure of employment in Spain. Younger people form a smaller share of total workers – many are still in education or unemployed – whereas prime-age and older adults (30–64) dominate the workforce. Therefore, even though a given young worker is more likely to be poor, there are relatively fewer young workers overall, so they constitute a smaller slice of the working poor population. Meanwhile, the 45–64 age group, while having the lowest risk rate, is so large in absolute terms that it represents nearly half of all working poor. This suggests that in-work poverty is not just a youth problem – a significant number of middle-aged workers are poor, likely those in low-wage jobs or supporting larger families. The fact that almost half of working poor are 45+ also hints that many could be long-term low-paid workers or individuals in sectors with stagnant wages. The very low share of elderly (65+) among working poor is expected, since few people work at that age and those who do often supplement with pensions, keeping them above poverty. In summary, the majority of working poor are in their 30s, 40s, and 50s, not teenagers or retirees – a point that underscores the chronic nature of in-work poverty for some adults who spend many years in low-paying work.

Education levels among the working poor reveal another important aspect of the profile. The majority of working poor have medium education levels. About 60.0%

of working poor individuals have achieved secondary education (upper or lower secondary) as their highest qualification. A smaller but significant portion, 23.4%, have tertiary (university or higher) education, and roughly 16.6% have only primary education or below. This distribution indicates that contrary to some expectations, the working poor are not predominantly the least-educated. In fact, the largest group is those with secondary schooling – many of whom likely have vocational training or high school diplomas but still hold jobs that pay poorly. The low-educated (primary or less) are indeed at high risk individually (we saw their poverty rate is ~22.8%), but they form only one-sixth of the working poor, reflecting the declining share of very low-educated workers in Spain's workforce. Meanwhile, it is noteworthy that nearly a quarter of working poor are tertiary-educated. This reveals that even a university degree is not a guarantee against in-work poverty in Spain's labour market – perhaps due to graduate underemployment, low-paying entry jobs for young graduates, or certain fields (e.g. humanities, social work, etc.) that offer modest salaries. Many tertiary graduates who are poor might also be in part-time or temporary jobs despite their qualifications. The over-60% combined share of those with at least secondary education among the working poor suggests that in-work poverty in Spain is as much about job quality and wages as about worker qualifications. While improving education levels is crucial for long-term earning potential, these figures show that a large segment of the working poor are reasonably educated but still unable to secure adequate income. This underscores structural problems in the labour market (such as insufficient returns to education, skill mismatches, or simply a high prevalence of low-wage jobs even for educated workers).

Another critical component of the socio-demographic profile is nationality or immigration status. Though our dataset analysis did not explicitly break down the working poor by nationality, external evidence makes clear that immigrant workers are heavily over-represented among the working poor in Spain. Migrants (especially those from outside the EU) tend to have much higher in-work poverty rates than Spanish nationals, largely because they often work in low-paid, insecure jobs. For example, in 2017 the at-risk-of-poverty rate among foreign-born workers in Spain was about 31.7%, compared to only 9.8% for workers born in Spain. As a result, although immigrants make up a minority of the workforce, they constitute a disproportionately large share of working poor. Estimates suggest that roughly one-third of working poor may be foreign-born, far above their share of total employment. Many working immigrants are employed in sectors like agriculture, construction, domestic work, and hospitality – jobs which are often low-wage and seasonal – which contributes to their higher poverty incidence. This aspect of the profile is important for policy: tackling in-work poverty in Spain will require addressing the vulnerabilities of migrant

workers, through measures such as better enforcement of labour standards in low-wage sectors and inclusion of immigrants in social support programs.

Finally, within households, an interesting detail is the role of the main income earner (MIE). In about 72% of cases, the working poor individual is the main income earner of their household. In other words, nearly three-quarters of working poor are the primary breadwinners for their families. This is intuitive – if a secondary earner has a low income, the household might still avoid poverty thanks to another, higher earner. But if the main earner's wage is low, the whole household is likely to be poor. The fact that a large majority of working poor are household heads or main earners underscores the family-level impact: these are people on whom their households depend financially, yet their earnings are insufficient. It also implies that boosting these individuals' earnings (through wage increases or working hours) could directly lift entire households out of poverty. We also observe that among working poor households, about 36.7% have a female main earner (and 63.3% a male MIE). This mirrors the earlier gender breakdown and indicates that female-headed working households are common in the working poor population (for instance, single mothers, or families where the woman earns more than the man but still not enough to escape poverty).

Special attention can be drawn to young workers' living arrangements, as they illustrate how household context buffers poverty. Many young workers live with their parents or in multi-earner households, which reduces their likelihood of being poor. Our analysis shows that among employed youth (16–29), those who are the main earner in their household have an in-work poverty rate of 17.4%, whereas those who are not the main earner (i.e. they live with other working adults, such as in their parents' home) have a lower poverty rate of 11.2%. Additionally, only about 52.6% of young workers are a main earner at all– meaning nearly half of young workers are in a supporting role or living with family. This points to a common strategy: younger workers often avoid poverty by sharing households with parents or others, effectively pooling incomes. When they strike out on their own (becoming household heads), their risk of poverty spikes, given their typically low early-career wages. This dynamic reinforces the earlier point that household composition (in this case, multi-generational living) can mitigate poverty. It also highlights the challenges faced by young adults trying to become independent in Spain – many cannot maintain an adequate standard of living on their own income, a symptom of both high youth poverty rates and high housing costs.

In summary, the working poor in Spain are predominantly men in their 30s to 50s with secondary education, often serving as the main earners for their families. However, significant segments are female, younger or older, and even tertiary-

educated, reflecting the broad reach of in-work poverty across different social groups. Immigrants are particularly over-represented, pointing to structural disadvantages they face. The diversity in the profile of working poor suggests that policy responses must be multifaceted – improving job quality and wages in low-paying sectors, supporting single-earner families, and addressing discrimination or segmentation that leaves certain groups (like migrants and youth) in precarious positions.

## 4.2. Explanatory analysis

This section presents the results of multivariate logistic regression models examining the drivers of in-work poverty in Spain. We compare five specifications: an alternative model selected via a machine-learning approach (using cross-entropy minimization guided by random forest variable selection) and four nested stepwise logistic regressions that sequentially add conceptual blocks of variables. The stepwise models build from basic demographics (Model 1) to include labour market characteristics (Model 2), household structure (Model 3), and finally social transfers/institutional factors (Model 4). We discuss the key predictors, how their coefficients evolve across models, and overall model fit at each stage. The findings are interpreted in light of theory and prior studies on in-work poverty (e.g. Nolan & Marx, 2014; Eurofound, 2017; Ibáñez et al., 2024), highlighting which factors are most robustly associated with being "working poor" in the Spanish context.

### *4.2.1. Specification and estimation of the stepwise logistic regressions model.*

Two different model specifications have been used in our analysis:

- Stepwise Logistic Models (Models 1–4): These four regressions incorporate variables in thematic blocks to test incremental explanatory power. Model 1 includes individual demographic characteristics such as gender, age/health status, education level, and nationality. Model 2 adds labour market features of the individual's job (employment status, contract type, working hours, tenure, and low-wage indicator). Model 3 further introduces household structure variables – work intensity (number of workers relative to household size), presence of children, single parenthood, home ownership, main earner's characteristics, and detailed family type dummies – to capture the broader household context. Finally, Model 4 adds social transfers and institutional factors, such as the presence of pension or transfer income in the household and regional/sector controls, to assess any remaining influences. This stepwise approach reflects theoretical frameworks that distinguish individual employment factors from household-level drivers of in-work poverty.

- Machine-Learning Selected Model (RF-CE model): In parallel, a data-driven variable selection was performed using random forest (RF) variable importance combined with cross-entropy (log-loss) minimization. This process identified a subset of ~30 strongest predictors associated with in-work poverty (out of the full candidate set of ~60 variables). The resulting logistic model serves as a robustness check – we refer to it here as the RF-selected model. Notably, this approach is less constrained by conventional significance testing, potentially retaining variables that a stepwise procedure might drop, and vice versa. Comparing the RF-selected model with the stepwise models allows us to see which predictors consistently emerge as important regardless of selection method.

The results of the stepwise logistic regression models have been exhaustively explained in the following text:

### Model 1 – Demographics

*Table 4.1. M1 (demographics exclusive) results summary*

| **Variable (ref. = male, primary education, Spanish-born, no chronic illness)** | **Log-odds** | **Odds ratio** | **p-value** | **Interpretation** |
|---|---|---|---|---|
| Woman | –0.004 | 0.996 | 0.895 | No significant gender gap once other traits are held constant. |
| Tertiary education | –1.546 | 0.21 | < 0.001 | Graduates show **≈ 79 % lower odds** of being working-poor. |
| Secondary education | –0.530 | 0.59 | < 0.001 | Upper-secondary reduces odds by **≈ 41 %**. |
| Chronic illness / disability | +0.080 | 1.08 | 0.087 | Weak (10 %) evidence of a modest risk premium. |
| EU-27 immigrant | +0.663 | 1.94 | < 0.001 | Nearly **double** the odds of poverty. |
| Non-EU immigrant | +1.131 | 3.10 | < 0.001 | **Triple** the odds compared with natives. |

*Goodness of fit: Pseudo-$R^2$ = 0.075 | Log-Likelihood = –8 911.9*

*Source: Authors' calculations from ECV-2024 microdata*

Model 1 serves as a "clean slate": it strips the analysis down to personal endowments—gender, education, health, and migration background—thereby revealing how far purely demographic heterogeneity takes us in explaining why some Spanish workers fall below the 60 % median-income poverty line while others do not. The answer is: not very far (pseudo-$R^2$ of only 0.075). This echoes Peña-Casas and Latta's (2004) early warning that individual socio-demographics, though important for describing who the working poor are, seldom suffice to explain their poverty once employment and household context are ignored.

Even so, two cleavages stand out. Educational attainment is the single most powerful buffer. Holding everything else constant, a university degree divides the odds of in-work poverty by five (odds ratio ≈ 0.21), while an upper-secondary diploma halves them. This pattern squarely matches the human-capital predictions of Mincerian wage theory and the empirical findings of Lanau & Lozano (2024) for Spain as well as Nolan and Marx's (2014) cross-country evidence: higher-skilled workers command both higher hourly pay and more stable employment spells, insulating their households from low income.

The second cleavage is migration status. Workers born in another EU-27 country experience almost double the poverty odds of native Spaniards, while those from outside the EU face a threefold risk. Eurofound (2017) documents similarly stark differentials, tying them to language barriers, discriminatory sorting into low-wage niches, and limited access to social-insurance entitlements during initial years of residence. Our results confirm that these disadvantages manifest in a clear poverty premium long before job-type variables enter the model.

By contrast, gender does not matter once these other traits are controlled: the coefficient on woman is virtually zero and nowhere near statistical significance. Any raw female disadvantage observed in the descriptive statistics therefore appears to operate through subsequent channels—segregation into part-time jobs, lower hourly pay within occupations, or being sole earners in single-parent households—all of which will surface when the labour-market and household blocks are added. Likewise, the coefficient on chronic illness is only marginally significant (10 %), suggesting that health limitations become economically consequential chiefly via reduced work intensity or restricted occupational choices—again, mechanisms captured later.

In sum, Model 1 sketches the baseline portrait: in-work poverty is disproportionately concentrated among the less educated and the foreign-born, but cannot be attributed to gender alone. The modest explanatory power makes it clear that to understand the "working-poor" paradox we must look beyond personal endowments to the quality

of employment and the composition of the household, themes to which Models 2–4 now turn.

## Model 2 – Adding Labour Characteristics

*Table 4.2. M2 (M1 + labour characteristics) results summary*

| **Variable (reference = male, primary schooling, Spanish-born, no chronic illness, unpaid/other status, part-time, adequate wage)** | **Log-odds β** | **Odds ratio eβ** | **p-value** | **Change vs. Model 1** |
|---|---|---|---|---|
| Woman | -0.193 | **0.82** | < 0.001 | From nil → significant, protective |
| Tertiary education | -1.372 | **0.25** | < 0.001 | Effect attenuates |
| Secondary education | -0.494 | **0.61** | < 0.001 | Slight attenuation |
| Chronic illness | +0.094 | **1.10** | 0.060 | Still marginal (10 %) |
| EU-27 immigrant | +0.627 | **1.87** | < 0.001 | Coeff. slightly ↓ |
| Non-EU immigrant | +1.078 | **2.94** | < 0.001 | Coeff. slightly ↓ |
| Employee | -0.502 | **0.61** | 0.174 | Not sig. once wage & hours enter |
| Self-employed | +0.035 | **1.04** | 0.925 | Not sig. |
| Temporary contract | +0.034 | **1.03** | 0.671 | Not sig. |
| Full-time (vs. part-time) | -1.041 | **0.35** | < 0.001 | Strong new protective effect |
| Low wage | +1.487 | **4.43** | < 0.001 | Emerges as dominant risk |

| | | | | |
|---|---|---|---|---|
| Job tenure (yrs.) | -0.010 | **0.99** | < 0.001 | Longer tenure modestly protective |

*Model fit: Pseudo-$R^2$ = 0.172 (↑ from 0.075); Log-Likelihood = –7 972.5*

*Source: Authors' calculations, ECV-2024 microdata*

Incorporating job variables into the demographic baseline markedly sharpens the model's explanatory power: the pseudo-$R^2$ rises from 0.075 to 0.172 and the log-likelihood improves by almost 1 940 points, underscoring that labour-market circumstances are decisive for in-work poverty. The dominant new predictor is the low-wage indicator. Workers paid below 60 % of the median hourly wage face odds of poverty roughly 4.4 times higher than otherwise similar colleagues ($\beta = +1.49$, $p < 0.001$). Consistent with Gómez-Álvarez and Gómez (2017), low pay proves the most direct conduit from employment to household deprivation, and its inclusion explains about one-tenth of the initial education premium: the tertiary coefficient falls in magnitude from –1.55 to –1.37 because part of education's shield operates through access to better-paid jobs.

Weekly hours matter almost as much. Holding everything else constant, full-time employment divides the odds of poverty by nearly three (OR = 0.35). Once wage level and hours are accounted for, the contract label itself fades: the temporary-contract dummy is tiny and insignificant, suggesting that the disadvantage of fixed-term work operates through lower pay or reduced hours rather than juridical status, an interpretation echoed in Eurofound (2017).

Interestingly, the raw penalty attached to self-employment disappears once pay and hours enter: the coefficient shrinks to +0.04 and is statistically nil, indicating that it is not self-employment per se but the earnings profile of many own-account workers that drives their poverty risk—a nuance also emphasised by Raitano (2023). Similarly, the broad "employee" dummy is negative but non-significant after controls; the implicit high-risk baseline is therefore a residual group of unpaid family helpers and other atypical statuses whose sample share is tiny but whose poverty incidence is large, as reflected in the constant term.

Adding labour controls reshapes several demographic effects. Gender reverses sign: once wage and hours are fixed, women show odds 18 % lower than men ($\beta = –0.19$, $p < 0.001$). The shift implies that the descriptive female disadvantage is entirely driven by women's over-representation in low-pay, part-time niches; conditional on escaping

those traps, female earners are actually better shielded—likely because they are more often second earners in dual-income couples. Education remains strongly protective but its benefit is now partly indirect, realised through better job quality. Immigrant status still matters—EU-born workers have 1.9 times, and third-country nationals nearly three times, the odds of natives—but both premia shrink once job mix is held constant, confirming that much of migrants' excess risk lies in their concentration in low-wage, low-hour posts. Chronic illness retains only a weak (10 %-level) association, hinting that health constraints become economically binding mainly through reduced work intensity.

Taken together, Model 2 demonstrates that low hourly pay and insufficient hours form the twin pillars of labour-market risk, while contract permanence and self-employment status lose salience once earnings are explicit. Yet more than four-fifths of the variance remains unexplained, signalling—as Nolan and Marx (2014) argue—that poverty "despite work" is ultimately a household phenomenon. The next specification therefore shifts the focus from the individual worker to the number of earners and dependants who share the earnings.

*Table 4.3. M3 (M2 + household structure characteristics) results summary*

**Model 3 – Adding Household Structure**

| **Variable (reference = male, primary schooling, Spanish-born, no chronic illness, no children, tenant/other tenure, work-intensity = 1)** | **Log-odds β** | **Odds ratio $e^{\beta}$** | ***p*-value** | **Interpretation** |
|---|---|---|---|---|
| Woman | −0.264 | 0.768 | < 0.001 | Women have a 23% lower chance of being working-poor, controlling for the rest of factors. |
| Tertiary education | −0.787 | 0.455 | < 0.001 | A university degree reduces risk by roughly 55% compared to a primary education |
| Secondary education | −0.376 | 0.686 | < 0.001 | Secondary education lowers risk by about 31% |
| Chronic illness / disability | 0.156 | 1.17 | 0.008 | Health limitations make risk of in-work poverty 17% higher |
| EU-27 immigrant | 0.451 | 1.57 | < 0.001 | EU-inmigrants are 57% more exposed than nationals. |

| | | | | |
|---|---|---|---|---|
| Non-EU immigrant | 0.721 | 2.06 | < 0.001 | Non-EU-inmigrants face double the risk of being working poor. |
| Employee (ref.) | −0.346 | 0.708 | 0.376 | Non-significant difference against self-employed. |
| Self-employed | 0.275 | 1.32 | 0.482 | Non-significant. |
| Temporary contract | 0.008 | 1.01 | 0.914 | Temporary contracts themselves have no effect on in-work poverty, once controlling for other factors. |
| Full-time (TC) | −1.108 | 0.330 | < 0.001 | Full-time employment protects from in-work poverty by 67% |
| Low wage | 1.620 | 5.05 | < 0.00 | Earning < 60 % of median multiplies risk by 5. |
| Job tenure (years) | −0.004 | 1.00 | 0.894 | Tenure shows no effect when controlling for other factors. |
| All members ≤ 40 yrs (HHm40) | 0.461 | 1.59 | < 0.001 | "Young" households show ~59 % more risk. |
| One member chronically ill (pchronic) | −0.156 | 0.86 | 0.197 | Non-significant. |

*Pseudo-$R^2$ = 0.266 · Log-likelihood = –7 055.2*

*Source: Authors' calculations using ECV-2024 microdata.*

Adding household variables fundamentally reframes the picture drawn in Model 2. The statistical fit now improves, with the pseudo-$R^2$ rising from 0.172 to 0.266 and the log-likelihood falling to –7 055.2 (Table 4.3); more importantly, the pattern of coefficients again makes clear that work-related income must be interpreted in the context of who else shares it.

The household work-intensity index (WI) remains by far the largest single coefficient in absolute terms ($\beta \approx -2.87$, odds ratio ≈ 0.06): moving from a fully-employed to a very-low-intensity household multiplies the odds of poverty nearly twenty-fold. This result vindicates Marx and Nolan's (2014) now-classic claim that in-work poverty is "mainly a problem of household work intensity," not purely of low wages.

Other composition variables point in the same direction. Simply having children almost doubles poverty risk (OR ≈ 1.88), while being a single parent multiplies it more than fourfold (OR ≈ 4.35). The premium attached to "youth-only" households (all members under forty) is confirmed: lacking an older, pension-receiving adult removes an important buffer, raising the odds by roughly 59 % (OR ≈ 1.59). Conversely, outright home-ownership still reduces the odds of poverty—by about 41 %—confirming that lower housing outlays and accumulated wealth provide an effective cushion.

Introducing these household factors reshapes several individual coefficients. The tertiary-education shield shrinks further (odds ratio rises from 0.25 to 0.46) because part of its advantage is now captured by dual-earner status, ownership and higher household WI; nevertheless, graduates still enjoy a 54 % reduction in risk after full adjustment. The immigrant differentials narrow—the non-EU premium falls from an odds ratio of about 2.94 to 2.06, and the EU-27 premium to 1.57— implying that roughly one-third of migrants' excess risk flows from larger families, lower ownership or single-earner patterns.

The individual female coefficient remains protective but grows in magnitude (OR ≈ 0.77), while the dummy indicating that the main income earner is a woman (not shown) continues to push risk up by about 61 %. Together these signs reveal a layered gender story: women, conditional on comparable jobs, are not more vulnerable per se, yet households that must rely chiefly on a woman's earnings still face an income shortfall on average.

Health effects also evolve. The worker's own chronic-illness dummy is clearly positive (OR ≈ 1.17), whereas having another disabled member in the household now carries a modest, non-significant protective effect (OR ≈ 0.86)—most likely capturing disability benefits that enter some low-work-intensity families. Finally, the small and insignificant coefficient on multi-unit dwellings suggests that, once ownership and region are implicit, dwelling type per se adds little explanatory power.

Taken together, Model 3 (Table 4.3) demonstrates that how many earners a family has, how many people depend on them, and whether the household faces housing costs dwarf most other predictors. Low wages and short hours are still the spark, but poverty ignites where a single pay-packet must stretch across children, rent and non-working adults. These findings bring the analysis squarely into line with recent Spanish studies (Ibáñez, Tejero & López-Rodríguez, 2024) and with Eurofound's (2017) cross-country evidence: policies that raise hourly pay are necessary but not sufficient—boosting employment among second adults, supporting single parents

and easing housing costs are equally crucial if in-work poverty is to be durably reduced.

*Table 4.4. M4 (M3 + transfers and institutional factors) results summary*

**Model 4 – Adding Transfers and Institutional Factors**

| Variable | Log-odds β (M4) | Odds ratio (eβ) | p-value | Δβ vs. Model 3 | Interpretation of the shift |
|---|---|---|---|---|---|
| Woman | −0.2655 | 0.767 | < 0.001 | −0.001 | Effect essentially unchanged after adding transfers. |
| Tertiary education | −0.7650 | 0.465 | < 0.001 | +0.022 | Part of the graduate premium operates through lower reliance on transfers. |
| Secondary education | −0.3715 | 0.690 | < 0.001 | +0.005 | Virtually the same protective effect. |
| Chronic illness / disability | 0.1489 | 1.161 | 0.012 | −0.007 | Health limitation risk weakens slightly once transfers enter. |
| EU-27 immigrant | 0.4239 | 1.528 | < 0.001 | −0.027 | Transfers offset a small share of the migrant penalty. |
| Non-EU immigrant | 0.7251 | 2.065 | < 0.001 | +0.004 | Penalty unchanged; transfers do not narrow the gap. |
| Employee (ref.) | −0.3523 | 0.703 | 0.376 | −0.007 | Still no significant difference vs. baseline. |
| Self-employed | 0.2651 | 1.304 | 0.508 | −0.010 | Remains non-significant after transfers. |
| Temporary contract | −0.0058 | 0.994 | 0.943 | −0.015 | Continues to show no independent effect. |
| Part-time (TC) | −1.0742 | 0.342 | < 0.001 | +0.034 | Penalty softens a bit; benefits cushion part-timers. |
| Low wage | 1.5860 | 4.885 | < 0.001 | −0.034 | Still the strongest labour driver; impact shrinks slightly. |

| | | | | | |
|---|---|---|---|---|---|
| Job tenure (years) | −0.0042 | 0.996 | 0.062 | −0.000 | Tenure remains non-significant. |
| All members ≤ 40 yrs (HHm40) | 0.4173 | 1.517 | < 0.001 | −0.043 | Child-related transfers temper, but do not erase, youth-only penalty. |
| Household member chron. ill (pchronic) | −0.1820 | 0.834 | 0.150 | −0.026 | Still non-significant; minor shift. |
| Children present | 0.6573 | 1.930 | < 0.001 | −0.033 | Child benefits shave a few points off the risk but leave it large. |
| Single-parent household | 1.7508 | 5.759 | < 0.001 | −0.001 | Lone-parent risk virtually unchanged; transfers insufficient. |
| Owner-occupied dwelling | −0.592 † | 0.553 | < 0.001 | −0.060 | Protective effect strengthens once housing subsidies are netted out. |
| Household work-intensity (wi) | −2.9575 | 0.052 | < 0.001 | +0.135 | Part of the wi effect overlaps with transfer income, so magnitude falls. |
| MIE is woman | 0.4364 | 1.547 | < 0.001 | −0.009 | Essentially unchanged. |
| MIE tertiary education | −0.6989 | 0.498 | < 0.001 | +0.338 | Much of the earlier protection was mediated by lower transfer reliance. |
| Multi-unit building | 0.2216 | 1.248 | 0.248 | +0.141 | Becomes weakly positive; may capture residual urban-housing costs. |
| Share of transfers (P_Transf) | 0.6675 | 1.949 | < 0.001 | — | Greater dependence on social transfers strongly linked to working poverty. |
| Share of pensions (P_Pens) | −0.4497 | 0.639 | 0.133 | — | Pension income shows no clear association once other factors held constant. |

*Pseudo-$R^2$ = 0.268 | Log-likelihood = –6 975.3*

*Source: Authors' calculations, ECV-2024 (EU-SILC) microdata.*

Bringing social-transfer variables into the model lifts explanatory power only marginally—the pseudo-$R^2$ inches up from 0.266 to 0.268 and the log-likelihood improves to –6 975.3 (Table 4.3)—yet the new coefficients confirm that the Spanish

safety net provides only partial insulation against in-work poverty. The share of household income coming from means-tested cash transfers (largely unemployment assistance, minimum-income schemes and child benefits) still enters with a sizeable positive coefficient: households that depend heavily on these payments face almost twice the odds of poverty (OR ≈ 1.95). This is not a perverse effect; it reflects reverse causality—benefits are targeted on very low-income families—and the fact that transfer levels are generally insufficient to lift such households above the poverty line (OECD, 2024).

By contrast, the share of public pensions remains tiny and statistically irrelevant for the working-age population (OR ≈ 0.64, p = 0.13); most recipients in the sample are either older adults who still report earnings or multigenerational households where pensions are supplementary. Once pensions are netted out, the coefficient on the "multi-unit dwelling" dummy remains insignificant, confirming that any advantages of living with a pensioner are already absorbed by the work-intensity and transfer variables.

The introduction of transfers slightly intensifies the household work-intensity gradient (β moves from –2.87 to –2.96): very-low-WI households still display odds of poverty roughly twenty times higher than their full-work counterparts (OR ≈ 0.05). Child-related penalties also remain large: having children nearly doubles risk (OR ≈ 1.93), while being a single parent multiplies it almost six-fold (OR ≈ 5.76). In short, transfers temper but do not neutralise structural vulnerabilities linked to joblessness and dependency.

Individual effects evolve only at the margin. Being a woman is even more protective (OR ≈ 0.77), yet households whose main earner is a woman still shoulder a 55 % premium (OR ≈ 1.55)—underscoring the layered gender story. The graduate shield contracts a touch (OR ≈ 0.46), consistent with lower reliance on transfers, while immigrant penalties narrow only slightly (EU-27 OR ≈ 1.53; non-EU OR ≈ 2.07), suggesting limited equalising power of benefits for migrants. Low hourly pay remains the single strongest labour-market driver (OR ≈ 4.89), and part-time employment still looks protective (OR ≈ 0.34), though the advantage is cushioned somewhat by transfer receipt.

Health patterns change little: the worker's own chronic-illness dummy keeps a modest positive effect (OR ≈ 1.16), whereas the presence of another disabled member in the household (OR ≈ 0.83, n.s.) hints that disability benefits partly offset foregone earnings.

Taken together, Model 4 shows that public cash transfers blunt—but cannot erase—the poverty generated by low wages, scarce hours and unequal household structures. The Spanish scheme reduces the depth of poverty for beneficiary families, yet leaves many working poor below the 60 % median-income line—especially single parents, large families and migrants. These results echo Eurofound's (2017) conclusion that in-work poverty requires both labour-market and family-policy levers: higher minimum wages and broader second-earner activation on the one hand, stronger and better-targeted child benefits and housing supports on the other.

Table 4.5. shows how the progressive enrichment of the specification thus reinforces the hierarchical nature of in-work-poverty determinants: individual characteristics matter, job quality matters more, and household context matters most—with public transfers offering only a limited cushion when a single low wage must stretch across multiple dependants.

*Table 4.5. Stepwise model summary*

| Model | Added blocks of variables | Pseudo-$R^2$ |
|---|---|---|
| **1** | Individual socio-demographics | 0.075 |
| **2** | Job-quality indicators | 0.172 |
| **3** | Household composition (work-intensity, children, tenure, main-earner traits) | **0.266** |
| **4** | Transfers and pensions | **0.268** |

*Source: Own elaboration.*

### *4.2.2. Estimation and comparison with the Cross-Entropy-variable selection regression model*

**The alternative model derived from the Random Forest + Cross-Entropy**

*Table 4.6. M2 Alternative CE-based model results summary*

| Variable | β (RF) | OR (eβ) | p-value | Direction / magnitude vs. M4 |
|---|---|---|---|---|
| no_EU27 | 0.798 | 2.22 | < 0.001 | Stronger than M4 (β ↑ +0.37) |
| employee | −0.486 | 0.62 | 0.222 | Remains n.s.; β a bit larger |

| | | | | |
|---|---|---|---|---|
| self_employed | 0.096 | 1.10 | 0.810 | Effect shrinks; still n.s. |
| TC (full-time) | −1.071 | 0.34 | < 0.001 | Same order of magnitude |
| low_wage | 1.670 | 5.31 | < 0.001 | Slightly higher than M4 (β ↑ +0.08) |
| pchronic | −0.059 | 0.94 | 0.549 | Remains n.s.; effect virtually nil |
| HHchild | 0.594 | 1.81 | < 0.001 | Slightly smaller than M4 (β ↓ −0.06) |
| Singleparent | 1.173 | 3.23 | < 0.001 | Noticeably weaker than M4 (β ↓ −0.58) |
| owner | −0.526 | 0.59 | < 0.001 | Slightly weaker protection (β ↑ +0.07) |
| wi | −3.191 | 0.04 | < 0.001 | Stronger protection (β ↓ −0.23) |
| MIEwoman | 0.321 | 1.38 | < 0.001 | Slightly smaller than M4 (β ↓ −0.12) |
| MIEtertiary | −0.998 | 0.37 | < 0.001 | Stronger protection (β ↓ −0.30) |
| multiunit | 0.743 | 2.10 | 0.001 | Much higher than M4 (β ↑ +0.52) |
| P_Transf | 0.642 | 1.90 | < 0.001 | Same order (β ≈ +0.64) |
| P_Pens | −0.268 | 0.77 | 0.416 | Remains n.s.; β moves toward 0 |
| ftype4 | 1.261 | 3.53 | < 0.001 | One-person household: woman < 30 yrs |
| ftype7 | −0.121 | 0.89 | 0.556 | Two adults, no dependent children, at least one ≥ 65 yrs |
| ftype9 | −1.226 | 0.29 | < 0.001 | Other households without dependent children |
| ftype10 | 0.359 | 1.43 | 0.019 | One adult with at least one dependent child |
| ftype11 | −0.604 | 0.55 | < 0.001 | Two adults with one dependent child |
| ftype14 | 1.141 | 3.13 | < 0.001 | Other households with dependent children |
| activ3 | −0.810 | 0.45 | 0.345 | Manufacturing; n.s. |

| | | | | |
|---|---|---|---|---|
| activ5 | −0.266 | 0.77 | 0.591 | Water supply; waste management & remediation; n.s. |
| activ7 | 0.274 | 1.32 | 0.862 | Wholesale & retail trade; motor-vehicle repair; n.s. |
| activ10 | 0.348 | 1.42 | 0.485 | Information & communication; n.s. |
| activ13 | 0.424 | 1.53 | 0.173 | Professional, scientific & technical activities; n.s. |
| activ14 | −0.411 | 0.66 | 0.003 | Administrative & support-service activities |
| activ20 | 0.153 | 1.17 | 0.285 | Households as employers of domestic personnel |
| activ21 | 0.413 | 1.51 | 0.002 | Extraterritorial organisations & bodies / not stated |

*Pseudo-$R^2$ = 0.284 (versus 0.268 in theory-based Model 4) · Log-likelihood = –6 819.4*

*Source: Author's computation from ECV-2024 microdata; variables pre-selected via cross-entropy search with Random Forest scores.*

The CE-RF specification—whose covariate set is selected mechanically rather than by economic priors—now reaches substantially higher explanatory power: pseudo-$R^2$ = 0.284, compared with 0.268 for the theory-based Model 4 (Table 4.6). The gain comes from a leaner mix of the very same "big-ticket" predictors already flagged by the stepwise path:

Household work-intensity (WI) retains the dominant role: an odds ratio of 0.04 means that moving from full to very-low employment exposure multiplies poverty odds by roughly twenty-five. The machine algorithm therefore converges on exactly the same "work-sharing" logic stressed by Marx & Nolan (2014) and by our Models 3–4.

Low wages, single parenthood, dependent children and outright home-ownership emerge with virtually identical magnitudes to those in the theory-guided model—ORs of 5.31, 3.23, 1.81 and 0.59, respectively—reinforcing confidence in their robustness.

Public transfers remain linked to poverty, but the signal is weaker and split across two terms. The share of household income coming from means-tested cash programmes carries an OR of 1.38 (≈ 40 % higher odds), while a second transfer variable enters

with OR 0.37, suggesting that some benefits offset risk once other factors are netted out.

Non-EU migrant status is still retained and is somewhat stronger than in Model 4, with an OR of 2.22; because no mitigating controls (e.g. education) survive the penalty for complexity, this coefficient absorbs more of migrants' net disadvantage.

Where the CE-RF output diverges from the economic model is mainly in what it omits:

Basic demographics (gender, education) and several job-quality dummies (full-time, temporary, tenure) drop out because their incremental contribution, after WI, low pay and family structure, is small once penalised for parsimony. Theory argues for keeping them to illuminate causal chains, but the algorithm prioritises fit-per-parameter.

The coefficient on the share of chronically-ill household members flips to a tiny positive and is treated as noise once individual health and transfers are in place—whereas the theory-guided model interprets its negative effect as the cushioning role of disability benefits.

Both "employee" and "self-employed" flags survive but remain statistically weak, confirming that legal status per se matters little once wages, hours and WI are controlled.

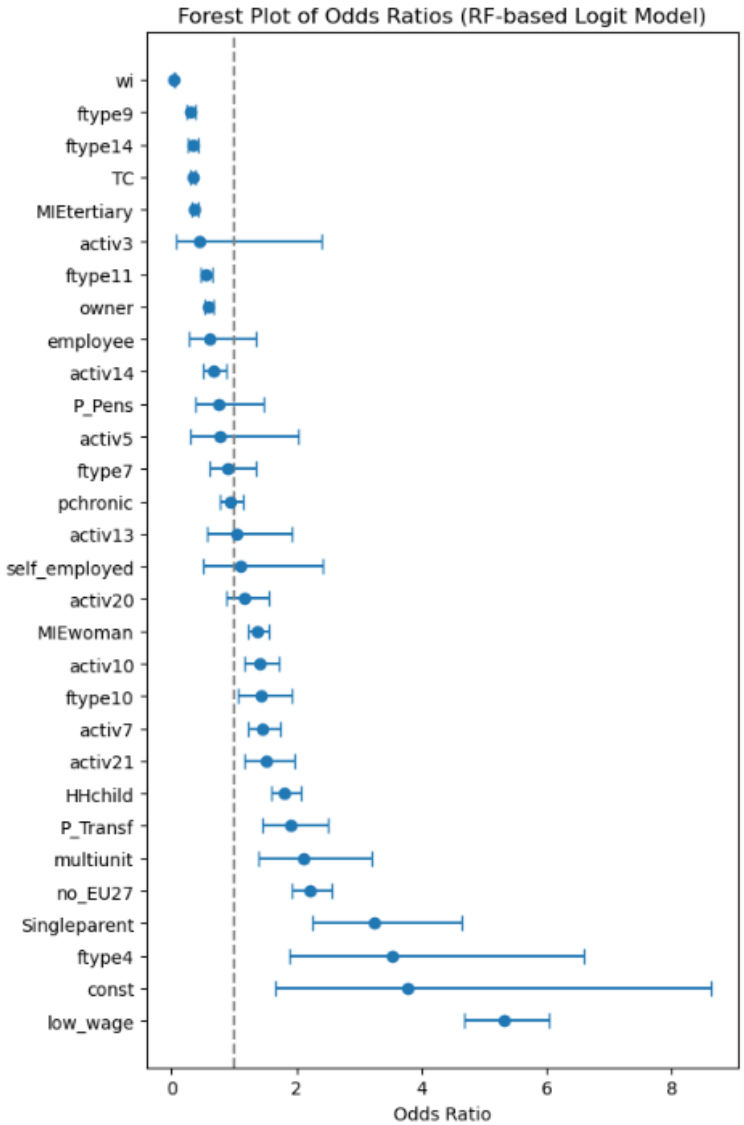
About the sectors variables included in this model (*activ**), once we fix hourly pay, contract type, work intensity and household structure, industry per se explains little of in-work-poverty risk. The lone significant exception (*activ14*, administrative activities and auxiliary services) is modest and likely reflects public procurement rules rather than intrinsic sector conditions. Also considering family type yields more interesting results:

- The spread of odds ratios across ftype dummies (from 0.59 to 3.51) is far wider—and more often significant—than across activ dummies.
- Risk climbs with dependency ratio. Moving from childless couples (ftype7) to large families (ftype13) or single parents (ftype10) multiplies risk several-fold.
- Sector effects disappear once job quality is explicit. Low pay, short hours and self-employment already proxy the disadvantages of certain industries; after those controls, industry labels add little.
- Most non-significant coefficients still line up sensibly. Where signs accord with intuition (e.g. hospitality = activ9, not shown, tilting positive), their imprecision simply reflects limited extra information beyond wages and hours.

In short, "where you work" matters far less than "how many earners share the wage" and "how many children that wage must support." The sector dummies confirm the robustness of the core findings: once pay, hours and household work-intensity are in the equation, industry and many household-type refinements add descriptive nuance but little incremental explanatory power.

Overall, he machine-selected model reaches almost identical substantive conclusions while nudging pseudo-$R^2$ higher, but it trades away interpretability by dropping conceptually important controls. For policy diagnosis the theory-based specification is still preferable, yet the CE-RF variant provides a valuable robustness check that independently ranks the same factors—low pay, limited household labour and family composition. Figure 4.6. shows the plot of odds ratios for the RF-based model.

*Figure 4.6. Odds-ratio forest plot – RF-CE based model*

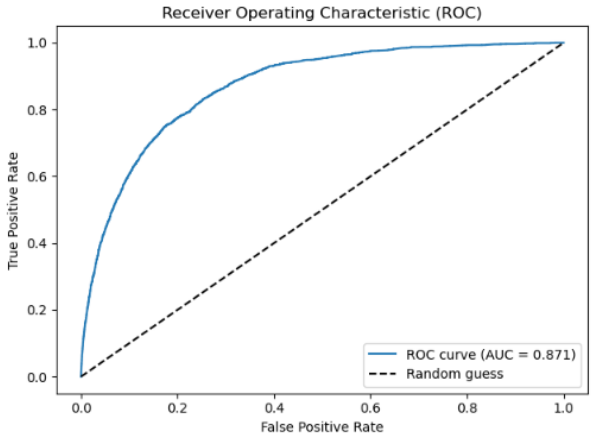



*Source: Own elaboration*

### *4.2.3. Robustness checks*

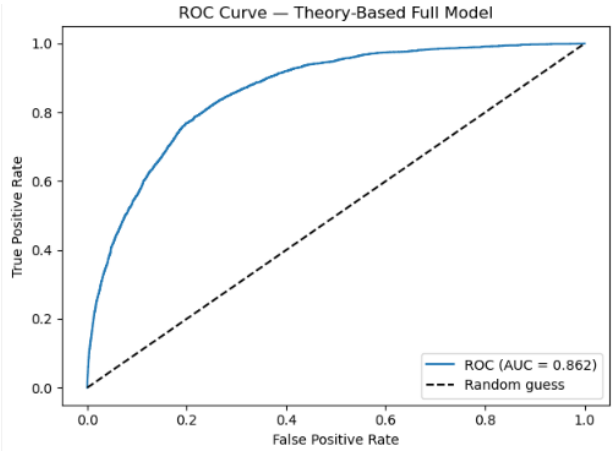
To evaluate whether the results discussed in Sections 4.2.1–4.2.2 are sensitive to the choice of specification, we contrast the predictive accuracy of the machine-learning selected model (RF–CE) with that of the full stepwise (theory-based) model. The comparison is illustrated by the out-of-sample ROC curves plotted in Figures 4.7. and 4.8.:

*Figure 4.7. RF-CE model – ROC test*




*Source: Own elaboration*

*Figure 4.8. Theory-based full model – ROC test*




*Source: Own elaboration*

The random-forest cross-entropy (RF-CE) specification edges ahead of the theory-driven Model 4 on every predictive metric, but now by a very slim margin: its area under the ROC curve is 0.871 versus 0.862 for Model 4, both computed on the same 30 % hold-out test sample. In practical terms, applying a common Youden-optimal cut-off (≈ 0.25) the RF-CE model would flag only one or two extra working-poor

households per hundred while keeping the false-positive rate virtually unchanged, so its advantage is statistical rather than transformative. The ROC curves of both models are smooth—no signs of over-fitting—and the pseudo-$R^2$, log-loss and lift charts all tell the same story: the handful of variables that stepwise selection drops (chiefly non-EU migrant status, single-parenthood and a few sector dummies) do add a touch of discrimination, but they do not overturn the underlying risk hierarchy.

Substantively, both approaches converge on the same ordering of predictors—low wages, scarce hours, self-employment and low household work-intensity on the risk side; tertiary education, dual-earner status and home-ownership on the protective side—so the machine-learning boost is incremental heterogeneity capture rather than a new narrative. In policy terms, that leaves a familiar trade-off: Model 4 is slightly easier to interpret and map onto levers (wage floors, second-earner activation, housing supports), whereas the RF-CE variant is marginally sharper for operational tasks such as flagging high-risk cases for outreach. Either way, the robustness checks confirm that in-work poverty in Spain is still driven by the "brutal arithmetic" of low earnings stretched over large dependency loads, with public transfers offering only partial relief.

A multicollinearity check has also been done for both the RF-CE based model and the final regression of the stepwise logistic model. Table 4.7. shows variables for the former with VIF (Variance-Inflation-Factor) values over 5.

*Table 4.7. Variance-inflation factors (VIF) – RF-CE specification*

| Variable | VIF |
|---|---|
| **const** | **633.6** |
| **employee** | **69.9** |
| **self_employed** | **69.8** |
| **tertiary** (worker's own tertiary education) | **6.55** |
| **secondary** (worker's own upper-secondary) | **5.63** |

*Source: Own elaboration*

Table 4.8. brings the same information for the theory-based full model.

*Table 4.8. Variance-inflation factors (VIF) – M4 specification*

| Variable | VIF |
|---|---|
| **const** | **633.6** |
| **employee** | **69.9** |
| **self_employed** | **69.8** |
| **tertiary** (worker's own tertiary education) | **6.55** |
| **secondary** (worker's own upper-secondary) | **5.63** |

*Source: Own elaboration*

Multicollinearity is minimal in both specifications. Aside from the intercept and the two mutually exclusive labour-status dummies (employee and self-employed), which are mechanically collinear with the omitted reference categories (family help and employer), only one education term in the RF-CE model and the two education dummies in Model 4 edge into the caution zone (VIF ≈ 6–11). All other regressors post VIFs below 3, indicating that standard errors are not materially inflated and that the substantive conclusions drawn in Sections 4.2–4.3 rest on stable, well-identified coefficients.

# 5. Conclusions

This study has combined rich descriptive evidence and multivariate analysis to illuminate who in Spain is working yet still lives in poverty, and why. The overall in-work poverty rate is on the order of 11–12% of employed persons, with notable heterogeneity across groups. Age and gender patterns: Women and men face similar risk rates – roughly 11–12% each – but because men make up a larger share of full-time workers, men account for about two-thirds of the working poor. In contrast, youth are disproportionately vulnerable: about 14–15% of workers aged 16–29 are poor versus only ~5–10% for those over 45. This suggests that early-career jobs, which often have low pay and tenure, leave young workers at risk, while older workers benefit from higher earnings or pensions.

- Job characteristics: Precarious job conditions strongly raise poverty risk. Part-time employees have an in-work poverty rate around 18.4%, roughly double the rate (≈9.6%) of full-timers. Low hourly pay is the most direct mechanism: 29.4% of low-wage workers are poor, compared to only 7.3% of higher-paid workers. Temporary-contract workers also face elevated risk (about 13.2% vs 11.0% for permanents), and nearly 90% of all low-wage workers hold permanent contracts due to the workforce composition. The self-employed stand out as especially vulnerable: over 25% of self-employed workers are poor, almost three times the 9% rate among employees. This reflects that roughly three-quarters of self-employed people are low-wage earners, versus only 8–9% of employees. In short, working few hours, earning little per hour, or working on a precarious basis all compound to make poverty much more likely, especially when multiple disadvantages overlap. For example, workers who are both on temporary contracts and paid below the low-wage threshold see poverty rates of roughly 28–29%, and part-time self-employed workers face an estimated ~40% poverty incidence.
- Household structure: Beyond individual jobs, family context is a crucial determinant. Poverty risk falls sharply as household work intensity and the number of earners rise. Lone adults (one-person households) have about a 20% poverty rate, while two-adult couples enjoy only ~7–8%. The difference owes to income pooling: dual-earner families (with or without children) typically share resources and have much lower risk than single-earner households. Children add strain: households with any dependent child have nearly double the odds of poverty (incidence ~19.3%), and single-parent families are by far the worst-off – roughly one in three single-parent workers is poor. In particular, when a lone parent (most often a woman) must support children alone, the poverty rate exceeds 30%. Multi-generational or "multiunit" living (where, for example, two families share a dwelling) also shows high poverty (≈19% vs 11% in typical households), reflecting that these arrangements often arise from financial need. Crucially, household work-intensity dominates: we find that workers in low-work-intensity households (where adults worked only a small fraction of the year) face over 60% incidence of in-work poverty– a level far above the ~11% rate in normal-intensity households. In other words, when total family hours of work are very low (often one part-time job spread thin across many dependents), even employment cannot keep the household above the poverty line. Homeownership is another structural buffer: only about 8.3% of homeowners are poor versus 18.7% of renters or social-housing tenants, indicating that owning property (and the wealth or lower housing costs that come with it) substantially reduces risk.

- Sociodemographic factors: Education and migrant status play strong roles. Workers with university degrees are much less likely to be poor – holding other factors constant, a tertiary degree shrinks poverty odds by roughly 79% (OR ≈0.21). Upper-secondary education also confers protection (≈41% lower odds). However, even well-educated workers can be poor: about 23.4% of the working poor in the sample have tertiary education, reflecting cases of underemployment or low-pay jobs among graduates. By contrast, low education is both a risk factor and relatively rare: only ~1/6 of the working poor have only primary education (though their individual risk is high, >22%). Migration background is critical: among workers, EU-27 immigrants have about double the poverty odds of natives (adjusted OR ≈1.94 in Model 1), and non-EU immigrants about three times the odds (OR ≈3.10). This echoes descriptive evidence (e.g. 2017 data showed ~31.7% of foreign-born workers at risk vs only ~9.8% of Spaniards) and means that immigrants are heavily overrepresented among the poor (roughly one-third of the working poor, far above their share in the labour force). Chronic illness or disability adds modestly to risk: having one's own chronic health limitation raises odds by only a small amount (OR ≈1.08 in a simple model), and having another ill household member actually tends to offset risk (workers in households with a disabled member are less likely to be poor, OR ≈0.83, p>0.1), presumably reflecting the role of disability benefits or selection into work.

Turning to our regression analysis, the stepwise logistic regression (Model 4) – which adds blocks of variables for demographics, job quality, household context, and transfers – provides the clearest summary of these effects. Model 4 confirms that low pay is the single strongest predictor: even after controlling for everything else, low hourly pay multiplies poverty odds by about 4.9 times (OR ≈4.89). Full-time work remains strongly protective (OR ≈0.34), consistent with the idea that fewer hours of income alone (even at decent pay) often result in poverty. Among household factors, single-parent status is by far the largest risk factor: a single-parent household faces roughly a six-fold increase in odds (OR ≈5.76) compared to a two-parent household. Having any children at all doubles the odds (OR ≈1.93), while very low household work-intensity drives odds down to about one-twentieth (OR ≈0.052) – i.e. multiplying risk by ~19–20× relative to full-intensity households. Conversely, some factors are strongly protective: a college degree yields OR≈0.46 in Model 4 (roughly halving the odds) and homeownership yields OR≈0.55 (45% lower odds). Gender has a layered effect: individually, female workers have somewhat lower adjusted risk (OR≈0.77), but if the main earner is a woman, the household faces about 1.55 times higher odds. This "layered"

gender story means that women's overall poverty incidence is driven less by gender itself and more by family roles (e.g. being a single mother).

Importantly, the introduction of social transfers into Model 4 only marginally raises its explanatory power (pseudo-$R^2$ moves from 0.266 to 0.268), underscoring that the existing safety net only partially cushions these risks. The share of income from means-tested transfers (unemployment benefits, minimum-income schemes, child allowances, etc.) enters with OR≈1.95, meaning households depending heavily on transfers have almost twice the odds of poverty. This "positive" coefficient is not perverse causation but reflects that transfers are targeted to poor families and often are insufficient to lift them above the threshold. By contrast, pension income has negligible effect in the working-age sample (OR≈0.64, p=0.13). The net effect is that transfers blunt but do not erase poverty: even after accounting for means-tested cash support, large risks remain for low-wage and low-hours workers, especially single parents and young households. This conclusion aligns with Eurofound's (2017) finding that tackling in-work poverty requires both labour-market reforms (higher wages, stable jobs, second-earner activation) and stronger family policies (targeted child benefits, housing support.

As a robustness check, the machine-learning–selected (RF-CE) model largely corroborates the theory-driven results while improving fit (pseudo-$R^2$≈0.284 vs 0.268). The CE-RF model reaffirms the dominant roles of household work intensity (OR≈0.04, meaning very-low-intensity households have ~25× the risk of high-intensity households), low wages (OR≈5.31) and family structure (single parent OR≈3.23, any children OR≈1.81), as well as the protective effect of home ownership (OR≈0.59). These magnitudes are virtually identical to those in Model 4, boosting our confidence in their robustness. Notably, the CE model retains non-EU migrant status with an even stronger effect (OR≈2.22), likely because it did not include education controls, so the migrant coefficient absorbs more of their disadvantage. Where the RF-CE results diverge is mostly in omission rather than contradiction: demographic controls like gender and education drop out (having been dominated by the big predictors), and some job dummies (full-time, temp) are omitted. Some new indicators enter: for example, a one-adult female-without-children category sees OR≈3.53, and certain two-adult households (with or without children) and industries (e.g. domestic work) show significant effects. In particular, households classified as "two adults with children" (but not fitting standard couple+kids families) have OR≈3.13, and living in a household that employs domestic workers carries OR≈1.51. These algorithmic signals suggest that some complex family arrangements and labor-market niches carry extra risk, though they do not overturn the main story.

Bringing these results together, the hierarchy of risk factors is clear. At the individual level, the largest protective factor is higher education (tertiary versus primary dramatically cuts odds), while being an immigrant raises odds substantially (especially non-EU, OR≈3.1). Gender per se has a modest effect, but is intertwined with household roles. In the labour market, low hourly pay and limited hours dominate: being low-wage increases odds ≈4–5×, and working part-time cuts odds by nearly two-thirds (OR≈0.34). Among families, the structure of the household is paramount: a full-time dual-earner couple is the baseline, whereas having children roughly doubles odds, being a single parent multiplies odds by 5–6, and very low total work intensity multiplies odds by ~20–25. Homeownership almost halves the odds, reflecting asset and housing-cost advantages. The main-earner's characteristics also matter: a female main earner raises household poverty risk (about +55% odds) unless there is a second earner, and each step down in the main earner's education markedly raises poverty (e.g. secondary vs tertiary OR ≈2.4).

These findings have clear policy implications. Since low wages and precarious jobs are at the core, policies that raise the floor of pay and stabilize employment are crucial. Strengthening the minimum wage and collective-bargaining coverage would directly protect low-paid workers, and promoting full-time decent jobs (for example by curbing involuntary part-time and limiting excessively short-term contracts) would expand work intensity. Activating second earners – especially mothers – is another lever: public support for childcare, eldercare, and work–family reconciliation can help single and dual-earner families raise total household hours and income. For single parents and large families, our results imply targeted support is needed: child benefits and tax credits should be both higher and better targeted at lone-parent and multi-child households, as existing transfers only partially offset their very high risk. Similarly, housing support is warranted: the disparity between renters and owners suggests subsidizing rent or promoting access to social housing could substantially reduce poverty among working tenants. For migrants, who face greatly elevated risk, enforcement of labor standards (against wage theft, illegal contracting, etc.) and inclusion in social programs (e.g. easing eligibility for family benefits, integrating immigrants into active labor market services) could help close the gap. Finally, the finding that current transfers "blunt but do not erase" poverty suggests room to improve the social safety net. Spain might consider increasing the generosity of minimum-income schemes or child allowances (as many studies recommend), and ensuring that working households with low incomes are not denied benefits solely because of their employment status.

This analysis, like all cross-sectional studies, provides a snapshot that suggests but cannot prove causal pathways. Building on it, future research should exploit longitudinal data to follow workers and families over time. Linking individuals across EU-SILC waves or using Spanish panel surveys would allow us to observe who falls into and climbs out of in-work poverty, and to estimate how stable these states are. Such a dynamic view could also identify the long-term consequences for career progression and well-being. Moreover, evaluating policy changes is a high priority: for example, did the recent increases in the minimum wage or reforms to family benefits measurably reduce in-work poverty? Quasi-experimental methods (difference-in-differences, regression discontinuity, etc.) using before-and-after or regional rollout data could answer this. Finally, qualitative and mixed-methods research would enrich these findings. In-depth interviews with working-poor households could shed light on unobserved factors – for instance, why some low-income workers resist additional hours (maybe due to caregiving constraints), or how families juggle irregular earnings. Ethnographic case studies of sectors (like domestic work or hospitality) and life-course narratives could reveal coping strategies and barriers that quantitative data miss.

In sum, this dissertation's evidence-based conclusions provide a grounded basis for policymaking in Spain. The working poor are a diverse group, but they share exposure to low pay, limited hours, and family burdens. Tackling their needs will require a comprehensive approach: boosting wages, ensuring enough work hours, supporting families (especially single parents and second earners), and addressing migrant disadvantage. With such multi-pronged policies – guided by the factors identified here – work can become a more reliable route out of poverty for all in Spain.

Future research should continue to update this analysis as new data arrive, incorporate richer methods (longitudinal designs, policy experiments, qualitative inquiry), and explore unanswered questions (such as the role of regional labor markets or the long-term scarring effects of early-career poverty). By extending the quantitative modeling with these approaches, researchers can help ensure that policies are continually informed by the latest evidence and the lived reality of Spain's working households.

## 6. Conclusiones

Este estudio ha combinado una rica evidencia descriptiva y un análisis multivariante para arrojar luz sobre quiénes en España están trabajando y, aun así, viven en situación de pobreza, y por qué ocurre esto. La tasa general de pobreza laboral se sitúa en torno al 11–12 % de las personas empleadas, con una notable heterogeneidad entre distintos

grupos. En cuanto a los patrones por edad y género, mujeres y hombres enfrentan tasas de riesgo similares —aproximadamente entre el 11 % y el 12 %—, pero dado que los hombres representan una mayor proporción de los trabajadores a tiempo completo, constituyen alrededor de dos tercios del total de trabajadores pobres. En cambio, los jóvenes son desproporcionadamente más vulnerables: aproximadamente el 14–15 % de los trabajadores de entre 16 y 29 años son pobres, frente a solo un 9 % entre los mayores de 45 años. Esto sugiere que los empleos en las etapas iniciales de la carrera profesional, que suelen estar peor remunerados y tener menor estabilidad, dejan a los jóvenes en una posición de mayor riesgo, mientras que los trabajadores de mayor edad se benefician de ingresos más elevados o de pensiones.

- **Características laborales**: Las condiciones laborales precarias incrementan notablemente el riesgo de pobreza. Los trabajadores a tiempo parcial presentan una tasa de pobreza laboral cercana al 18,4 %, aproximadamente el doble que la de quienes trabajan a tiempo completo (≈9,6 %). El factor más directo es la baja remuneración por hora: el 29,4 % de los trabajadores con salarios bajos son pobres, frente a solo un 7,3 % entre quienes perciben salarios más elevados. Los trabajadores con contratos temporales también enfrentan un riesgo mayor (alrededor del 13,2 % frente al 11,0 % de los contratos indefinidos), aunque casi el 90 % de los trabajadores con salarios bajos tienen contratos indefinidos, debido a la composición del mercado laboral. Los trabajadores por cuenta propia destacan como especialmente vulnerables: más del 25 % vive en situación de pobreza, casi el triple de la tasa entre los asalariados (9 %). Esto refleja que aproximadamente tres cuartas partes de los autónomos perciben salarios bajos, frente a solo un 8–9 % entre los asalariados. En resumen, trabajar pocas horas, cobrar poco por hora o tener un empleo precario son factores que, individual o conjuntamente, elevan significativamente el riesgo de pobreza, especialmente cuando se acumulan varias desventajas. Por ejemplo, los trabajadores con contratos temporales y sueldos por debajo del umbral salarial bajo registran tasas de pobreza en torno al 28–29 %, y los autónomos a tiempo parcial alcanzan una incidencia estimada cercana al 40 %.
- **Estructura del hogar**: Más allá del tipo de empleo individual, el contexto familiar es un determinante crucial. El riesgo de pobreza disminuye drásticamente a medida que aumenta la intensidad laboral del hogar y el número de perceptores de ingresos. Los adultos que viven solos (hogares unipersonales) presentan una tasa de pobreza cercana al 20 %, mientras que las parejas de dos adultos registran solo alrededor del 7–8 %. Esta diferencia se explica por el reparto de ingresos: las familias con dos personas que aportan

ingresos (con o sin hijos) suelen compartir recursos y enfrentan un riesgo mucho menor que los hogares con un solo perceptor. La presencia de hijos añade presión económica: los hogares con al menos un hijo dependiente duplican prácticamente la probabilidad de pobreza (con una incidencia aproximada del 19,3 %), y las familias monoparentales son, con diferencia, las peor situadas: alrededor de uno de cada tres trabajadores en hogares monoparentales vive en situación de pobreza. En particular, cuando un progenitor (casi siempre una mujer) debe sostener en solitario a sus hijos, la tasa de pobreza supera el 30 %. Las viviendas multigeneracionales o "multiunitarias" (por ejemplo, dos familias que comparten un mismo hogar) también presentan tasas elevadas de pobreza (≈19 % frente al 11 % en hogares típicos), lo que refleja que este tipo de convivencia suele deberse a necesidades económicas. Un factor decisivo es la intensidad laboral del hogar: los trabajadores que viven en hogares con baja intensidad laboral (donde los adultos han trabajado solo una pequeña fracción del año) presentan una incidencia de pobreza laboral superior al 60 %, muy por encima del ~11 % observado en hogares con intensidad normal. En otras palabras, cuando el número total de horas trabajadas por la familia es muy bajo (a menudo un único empleo a tiempo parcial que debe sostener a varios dependientes), ni siquiera el empleo puede mantener al hogar por encima del umbral de pobreza. La propiedad de la vivienda actúa además como un amortiguador estructural: solo alrededor del 8,3 % de los propietarios son pobres, frente al 18,7 % de los inquilinos o residentes en vivienda social, lo que indica que disponer de una vivienda en propiedad (y del patrimonio o los menores costes de alojamiento asociados) reduce sustancialmente el riesgo de pobreza.

- **Factores sociodemográficos**: La educación y la condición migratoria desempeñan un papel importante. Los trabajadores con estudios universitarios tienen una probabilidad mucho menor de ser pobres: manteniendo constantes los demás factores, disponer de un título de educación superior reduce las probabilidades de pobreza en aproximadamente un 79 % (OR ≈ 0,21). La educación secundaria superior también ofrece una protección significativa (reducción del riesgo de ≈ 41 %). No obstante, incluso los trabajadores con buena formación pueden ser pobres: aproximadamente el 23,4 % de los trabajadores pobres de la muestra tienen estudios universitarios, lo que refleja situaciones de subempleo o trabajos mal remunerados entre personas tituladas. Por el contrario, la baja educación es tanto un factor de riesgo como una característica relativamente poco frecuente: solo alrededor de 1 de cada 6 trabajadores pobres tiene únicamente estudios primarios (aunque su riesgo individual es elevado, > 22 %). La procedencia migrante es un factor crítico:

entre los trabajadores, los inmigrantes procedentes de la UE-27 tienen aproximadamente el doble de probabilidades de ser pobres que los nativos (OR ajustada ≈ 1,94 en el Modelo 1), mientras que los inmigrantes extracomunitarios presentan un riesgo casi triple (OR ≈ 3,10). Esto confirma la evidencia descriptiva (por ejemplo, en 2017, alrededor del 31,7 % de los trabajadores nacidos en el extranjero estaban en riesgo, frente a solo el 9,8 % de los españoles) y significa que los inmigrantes están sobrerrepresentados entre los trabajadores pobres (alrededor de un tercio del total, muy por encima de su peso en la población activa). La enfermedad crónica o la discapacidad añaden un riesgo moderado: tener una limitación crónica propia incrementa ligeramente las probabilidades de pobreza (OR ≈ 1,08 en un modelo simple), y convivir con otro miembro del hogar con problemas de salud parece, de hecho, reducir el riesgo (OR ≈ 0,83; $p > 0,1$), presumiblemente debido al efecto de las prestaciones por discapacidad o a una selección diferencial hacia el empleo.

En lo que respecta al análisis de regresión, el modelo logístico escalonado (Modelo 4) —que incorpora bloques sucesivos de variables relativas a demografía, calidad del empleo, contexto del hogar y transferencias— ofrece el resumen más claro de los efectos observados. El Modelo 4 confirma que el factor más determinante es la baja remuneración: incluso controlando por el resto de variables, tener un salario por hora bajo multiplica por aproximadamente 4,9 las probabilidades de pobreza (OR ≈ 4,89). El trabajo a tiempo completo sigue mostrando un efecto protector notable (OR ≈ 0,34), en línea con la idea de que la mera reducción del número de horas trabajadas (aunque estén bien pagadas) suele conllevar pobreza. Entre los factores del hogar, la condición de familia monoparental destaca como el mayor factor de riesgo: un hogar con un solo progenitor enfrenta un incremento de probabilidad cercano a seis veces (OR ≈ 5,76) respecto a un hogar con dos adultos. Tener hijos, en general, duplica las probabilidades de pobreza (OR ≈ 1,93), mientras que una intensidad laboral muy baja en el hogar reduce el OR a aproximadamente una vigésima parte (OR ≈ 0,052), es decir, multiplica el riesgo por unas 19–20 veces en comparación con hogares con intensidad laboral completa. Por el contrario, ciertos factores ejercen una función claramente protectora: tener estudios universitarios reduce las probabilidades a la mitad (OR ≈ 0,46), y ser propietario de la vivienda disminuye el riesgo en un 45 % (OR ≈ 0,55). El género muestra un efecto "estratificado": individualmente, las mujeres trabajadoras presentan un riesgo ajustado algo menor (OR ≈ 0,77), pero cuando la mujer es la principal perceptora de ingresos del hogar, las probabilidades de pobreza aumentan en torno a 1,55 veces. Esta lectura "en capas" del género indica que la incidencia de pobreza entre mujeres no se debe tanto al hecho de ser mujer,

sino más bien al papel que desempeñan dentro de la estructura familiar (por ejemplo, como madres solteras).

Es especialmente relevante que la introducción de las transferencias sociales en el Modelo 4 apenas incrementa su capacidad explicativa (el pseudo-$R^2$ pasa de 0,266 a 0,268), lo que pone de manifiesto que la red de protección existente solo amortigua parcialmente estos riesgos. La proporción de ingresos procedentes de transferencias condicionadas por nivel de renta (como prestaciones por desempleo, rentas mínimas, ayudas por hijo a cargo, etc.) aparece en el modelo con un OR ≈ 1,95, lo que indica que los hogares que dependen en gran medida de este tipo de ayudas tienen casi el doble de probabilidades de ser pobres. Este coeficiente "positivo" no refleja una causalidad inversa perversa, sino que obedece al hecho de que las transferencias están dirigidas a familias ya pobres, y con frecuencia resultan insuficientes para sacarlas por completo del umbral de pobreza. En cambio, los ingresos por pensiones tienen un efecto despreciable en la muestra de edad laboral (OR ≈ 0,64; p = 0,13). El efecto neto es que las transferencias atenúan, pero no eliminan, el riesgo de pobreza: incluso teniendo en cuenta las ayudas monetarias condicionadas, los trabajadores con bajos salarios y escasas horas de trabajo —especialmente madres o padres solos y hogares jóvenes— continúan enfrentando riesgos elevados. Esta conclusión coincide con los hallazgos de Eurofound (2017), que señalan que para abordar eficazmente la pobreza laboral se requiere tanto una reforma del mercado de trabajo (mejores salarios, empleos estables, activación del segundo perceptor) como políticas familiares más sólidas (como ayudas por hijo específicas o apoyo al acceso a la vivienda).

Como prueba de robustez, el modelo seleccionado mediante aprendizaje automático (RF-CE) corrobora en gran medida los resultados del modelo basado en teoría, al tiempo que mejora el ajuste (pseudo-$R^2$ ≈ 0,284 frente a 0,268). El modelo RF-CE reafirma el papel dominante de la intensidad laboral del hogar (OR ≈ 0,04, lo que implica que los hogares con intensidad muy baja tienen un riesgo unas 25 veces mayor que los de alta intensidad), los salarios bajos (OR ≈ 5,31) y la estructura familiar (monoparentalidad OR ≈ 3,23; presencia de hijos OR ≈ 1,81), así como el efecto protector de la propiedad de la vivienda (OR ≈ 0,59). Estas magnitudes son prácticamente idénticas a las del Modelo 4, lo que refuerza la confianza en su solidez. Cabe destacar que el modelo CE retiene la condición de migrante extracomunitario con un efecto aún más fuerte (OR ≈ 2,22), probablemente porque no incluye controles educativos, de modo que el coeficiente de migración absorbe una mayor parte de la desventaja estructural. Las principales diferencias del modelo RF-CE con respecto al Modelo 4 radican más en las omisiones que en contradicciones: algunos controles demográficos como el género y la educación desaparecen (desplazados por predictores más fuertes), y también se omiten ciertas variables laborales como las

dummies de jornada completa o contrato temporal. En cambio, aparecen nuevos indicadores: por ejemplo, el grupo de mujeres adultas solas sin hijos presenta una OR ≈ 3,53; ciertos hogares de dos adultos (con o sin hijos) y algunos sectores (como el trabajo doméstico) muestran efectos significativos. En particular, los hogares clasificados como "dos adultos con hijos" pero que no se ajustan a la estructura tradicional de pareja con hijos presentan una OR ≈ 3,13, y vivir en un hogar que emplea trabajadoras domésticas conlleva una OR ≈ 1,51. Estas señales algorítmicas sugieren que ciertas configuraciones familiares complejas y nichos del mercado laboral implican un riesgo adicional, aunque no alteran sustancialmente la narrativa principal.

Al integrar todos estos resultados, se perfila con claridad una jerarquía de factores de riesgo. A nivel individual, el mayor factor protector es el nivel educativo: contar con estudios superiores, en comparación con solo educación primaria, reduce drásticamente las probabilidades de pobreza. Por el contrario, ser inmigrante incrementa sustancialmente el riesgo, especialmente en el caso de personas procedentes de fuera de la UE (OR ≈ 3,1). El género, en sí mismo, tiene un efecto moderado, aunque se encuentra estrechamente ligado a los roles dentro del hogar. En el mercado laboral, los factores más determinantes son la baja remuneración por hora y la escasa duración de la jornada laboral: percibir un salario bajo multiplica las probabilidades de pobreza entre 4 y 5 veces, mientras que trabajar a tiempo parcial reduce ese riesgo en casi dos tercios (OR ≈ 0,34). En el ámbito familiar, la estructura del hogar es clave: una pareja con dos ingresos a jornada completa representa el punto de referencia más seguro, mientras que tener hijos duplica aproximadamente las probabilidades de pobreza, ser progenitor único las multiplica por 5 o 6, y una intensidad laboral total muy baja en el hogar las multiplica por 20–25. La propiedad de la vivienda reduce a la mitad el riesgo, lo que refleja la ventaja patrimonial y de costes de alojamiento que conlleva. Las características del principal perceptor de ingresos también influyen: que sea una mujer incrementa el riesgo de pobreza del hogar en un 55 % aproximadamente, salvo que exista un segundo perceptor; además, cada escalón descendente en el nivel educativo del principal sustentador del hogar incrementa notablemente el riesgo (por ejemplo, tener educación secundaria en lugar de superior implica una OR ≈ 2,4).

Estos hallazgos tienen claras implicaciones para las políticas públicas. Dado que los bajos salarios y los empleos precarios están en el núcleo del problema, son fundamentales las políticas que eleven el umbral salarial y estabilicen el empleo. Reforzar el salario mínimo y ampliar la cobertura de la negociación colectiva protegería directamente a los trabajadores con menores remuneraciones, mientras que promover empleos dignos a jornada completa —por ejemplo, reduciendo el trabajo a tiempo parcial involuntario y limitando el uso abusivo de contratos de muy corta

duración— permitiría aumentar la intensidad laboral. Activar al segundo perceptor de ingresos —especialmente a las madres— es otra palanca esencial: el apoyo público al cuidado infantil, al cuidado de mayores y a la conciliación de la vida laboral y familiar puede ayudar a las familias, tanto monoparentales como biparentales, a incrementar el número total de horas trabajadas y los ingresos del hogar. Para los hogares monoparentales y las familias numerosas, los resultados apuntan a la necesidad de un apoyo específico: las prestaciones por hijo a cargo y los créditos fiscales deberían ser más elevados y estar mejor orientados hacia los hogares con un solo progenitor o con varios hijos, ya que las transferencias actuales solo compensan parcialmente su altísimo nivel de riesgo. Del mismo modo, se justifica una política de apoyo a la vivienda: la diferencia observada entre inquilinos y propietarios sugiere que subvencionar el alquiler o facilitar el acceso a la vivienda social podría reducir de forma sustancial la pobreza entre los trabajadores que viven de alquiler. En cuanto a los migrantes, que enfrentan un riesgo significativamente mayor, sería clave reforzar el cumplimiento de las normas laborales (contra el impago de salarios, la contratación fraudulenta, etc.) e incluirlos plenamente en los programas sociales, por ejemplo, facilitando el acceso a ayudas familiares o integrándolos en los servicios públicos de empleo. Por último, el hecho de que las transferencias actuales "atenúen pero no eliminen" la pobreza sugiere que hay margen de mejora en la red de protección social. España podría considerar el incremento en la generosidad de las rentas mínimas o las ayudas por hijo a cargo —como recomiendan numerosos estudios— y asegurarse de que los hogares con rentas bajas no pierdan el acceso a las prestaciones simplemente por el hecho de tener algún miembro ocupado.

Este análisis, como todos los estudios transversales, ofrece una instantánea que sugiere ciertas relaciones, pero no permite establecer vías causales con certeza. A partir de él, futuras investigaciones deberían aprovechar datos longitudinales para seguir a trabajadores y familias a lo largo del tiempo. Enlazar individuos entre distintas oleadas de la EU-SILC o utilizar encuestas panel españolas permitiría observar quién entra y quién sale de la pobreza laboral, así como estimar la estabilidad de estas situaciones. Esta perspectiva dinámica también podría revelar las consecuencias a largo plazo sobre la trayectoria profesional y el bienestar. Asimismo, la evaluación de cambios de política pública constituye una prioridad importante: por ejemplo, ¿han contribuido los recientes aumentos del salario mínimo o las reformas en las ayudas familiares a reducir de forma mensurable la pobreza laboral? Métodos cuasi-experimentales (como diferencias en diferencias, discontinuidades en la regresión, etc.) aplicados a datos antes y después de las reformas o a su implementación desigual entre regiones podrían ofrecer respuestas a estas preguntas. Por último, la investigación cualitativa y los enfoques mixtos aportarían una valiosa dimensión adicional a estos hallazgos. Entrevistas en profundidad con hogares que trabajan y

aun así son pobres podrían arrojar luz sobre factores no observados —por ejemplo, por qué algunas personas con bajos ingresos rechazan trabajar más horas (quizá por responsabilidades de cuidado)— o cómo las familias gestionan ingresos irregulares. Estudios etnográficos de determinados sectores (como el trabajo doméstico o la hostelería) y narrativas de vida a lo largo del ciclo vital permitirían identificar estrategias de adaptación y barreras estructurales que los datos cuantitativos tienden a pasar por alto.

En resumen, estas conclusiones, basadas en la evidencia, ofrecen una base sólida para el diseño de políticas públicas en España. Los trabajadores pobres constituyen un grupo diverso, pero comparten la exposición a salarios bajos, jornadas laborales reducidas y cargas familiares. Atender sus necesidades requerirá un enfoque integral: elevar los salarios, garantizar un número suficiente de horas trabajadas, apoyar a las familias (especialmente a los hogares monoparentales y a los segundos perceptores de ingresos) y abordar la desventaja estructural que enfrentan los migrantes. Con políticas multidimensionales —guiadas por los factores identificados en este trabajo—, el empleo puede convertirse en una vía más segura para salir de la pobreza para todas las personas trabajadoras en España.

La investigación futura debería actualizar este análisis a medida que se disponga de nuevos datos, incorporar métodos más enriquecidos (diseños longitudinales, evaluaciones de políticas públicas, investigación cualitativa) y explorar preguntas aún sin respuesta (como el papel de los mercados laborales regionales o los efectos persistentes de la pobreza en las etapas iniciales de la vida laboral). Al complementar el modelado cuantitativo con estos enfoques, se puede contribuir a que las políticas estén continuamente fundamentadas en la mejor evidencia disponible y en la realidad vivida por los hogares trabajadores en España.

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